

Dittert's Conjecture in Dimensions Six through Fifteen: An Exact Computer-Assisted Proof

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Abstract

Let K_n be the simplex of nonnegative $n \times n$ matrices whose entries sum to n . If $A \in K_n$ has row sums $r_1, \dots, r_n$ and column sums $c_1, \dots, c_n$, define

$$\Phi(A) = \prod_{i=1}^n r_i + \prod_{j=1}^n c_j - \text{per}(A).$$

Dittert's conjecture asserts that Φ is uniquely maximized by the uniform matrix $U_n = n^{-1}J_n$, with maximum $2 - n!/n^n$. We prove the conjecture for every $6 \leq n \leq 15$. A Hall-deficit dilation converts a hypothetical boundary maximizer into a doubly stochastic matrix with a forced zero pattern. Stable-polynomial capacity yields support-sensitive permanent bounds and entropy refinements of the sharp rectangular-zero-block inequality. A complementary-minor entropy theorem resolves the one-unit Hall cores in dimension seven. Dimension six requires a broader core-marginal entropy estimate, an exact core-scaling stationarity identity, and a pairwise smoothing reduction to finitely many four-variable rational polynomials. Their positivity, together with the remaining scalar inequalities, is certified by exact rational Bernstein subdivision. An independent pairwise-averaging argument proves that every entrywise positive global maximizer is uniform. The finite part is verified by explicit rational certificate checkers whose success conditions remain active under every Python optimization mode. The source, independent audit implementations, hashes, and machine-readable verification manifest are supplied with the paper.

Keywords. Permanent; Dittert conjecture; doubly stochastic matrix; stable polynomial; polynomial capacity; Hall condition; Birkhoff polytope.

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1 Introduction

For an $n \times n$ nonnegative matrix $A = (a_{ij})$, let r_i and c_j denote its row and column sums. Dittert proposed the extremal problem of maximizing

$$\Phi(A) = \prod_i r_i + \prod_j c_j - \text{per}(A)$$

on the simplex K_n of matrices whose entries have total sum n . The conjectured unique maximizer is the uniform matrix $U_n = n^{-1}J_n$. On the doubly stochastic polytope the two product terms equal one, so Dittert's assertion reduces to the van der Waerden permanent theorem of Egorychev and Falikman [2, 3]. The conjecture was recorded by Minc [9] and later by Zhan [13].

Sinkhorn proved the case $n = 2$ [11], and Hwang proved the case $n = 3$ [6]; Hwang had earlier shown that an entrywise positive global maximizer must be uniform [5]. Knopp and Sinkhorn studied permanent lower bounds for doubly stochastic matrices with a zero [8], and Cheon and Wanless developed boundary-exclusion methods and other structural results [1]. Udayan and Somasundaram proved that every Φ -maximizing matrix on K_4 is fully indecomposable [12]. Recent preprints of Pang and Kafidov claim the conjecture for $n \geq 17$ and $n = 16$, respectively [10, 7]. These historical statements are not logical inputs to our proof. The present paper proves the ten dimensions

$$6 \leq n \leq 15.$$

The proof separates the interior and boundary of K_n . The interior classification is dimension-free: pairwise averaging is exactly flat at a positive global maximizer, whereas the uniform matrix is a strict local maximizer. Cyclic averaging and a backward strictness argument force every positive global maximizer to be uniform.

For a boundary maximizer, the row- and column-product losses share one deficit budget. Relative entropy controls subset sums, and a max-flow/min-cut criterion shows that a scalar dilation dominates a doubly stochastic matrix B . At a critical cut, B contains a rectangular zero block. Stable-polynomial capacity gives permanent lower bounds adapted to that support.

Dimensions 8, 9, 10 require a second-order scalar comparison and a separate analysis of one-sided Hall cuts. Dimensions 11, 12, 13, 14 are handled by first-order margins, apart from one deficient-column configuration in dimension 11, and dimension 15 admits a thin-thick dichotomy. Dimension seven is qualitatively different: three critical cuts with $n - a - b = 1$ survive the scalar method. A geometric-mean entropy inequality for complementary permanent minors strengthens the sharp zero-block bound on these one-unit cores. Stationarity and conditional product energy then reduce the remaining cases to three exact two-variable polynomial alternatives.

Dimension six requires one further level of structure. Critical cores with $n - a - b \geq 2$ can survive all scalar comparisons. We prove a core-marginal entropy refinement of the rectangular-zero-block bound that records both normalized marginal distributions of the Hall core. An exact core-scaling identity couples this entropy to the row and column products. A pairwise smoothing lemma reduces each side to a uniform distribution on its support and two levels of row or column sums. The six Hall block types then become 98 explicit four-variable rational polynomials. One-sided cuts are excluded by retaining the exact complementary column-product loss.

The finite positivity statements are verified by tensor-product Bernstein subdivision with exact rational coefficients. This is a computer-assisted proof component, not a floating-point experiment: every accepted box has strictly positive Bernstein coefficients, and every subdivision is exact de Casteljau averaging. The dimension-seven-through-fifteen checker uses only the Python standard library. The dimension-six generator uses `sympy.Rational` only to expand the displayed rational formulas; the Bernstein transformation, subdivision, and every correctness decision use exact integers and `fractions.Fraction`. Independent implementations reconstruct the 98 four-variable certificates, the ten one-variable certificates, and the three dimension-seven trees. The complete source code, a machine-readable manifest, and source hashes reproduce every certificate.

Two ingredients often taken from the literature are proved here rather than used as black boxes. We reconstruct the derivative-capacity mechanism associated with Gurvits [4], deriving all permanent bounds needed below, and we give an independent proof of Hwang’s positive-maximizer conclusion. We make no claim here about the complete status of dimensions outside $6 \leq n \leq 15$: the cited $n = 4$, $n = 16$, and $n \geq 17$ literature has different publication status and scope, and none of it is needed for our main theorem.

Proof architecture and certificate scale

Table 1 records where analytic structure ends and finite exact verification begins.

Dimensions	Principal analytic mechanism	Exact finite certificate
11–14	first-order Hall margins; one deficient-column refinement for $n = 11$	35, 42, 49, 56 rational cases
15	thin–thick support dichotomy	two rational margins and six integer inequalities
8–10	second-order Hall margins and one-sided entropy	12, 16, 20 rational Hall cases plus scalar certificates
7	complementary-minor entropy and single-bridge stationarity	three bivariate Bernstein trees
6	full core-marginal entropy, core stationarity, and two-level smoothing	98 four-variable and ten univariate Bernstein certificates

Table 1: Dimension-by-dimension analytic mechanism and exact certificate size.

The logical dependency is acyclic. Its boundary branch is

$$\boxed{\text{stable capacity}} \implies \boxed{\text{support/entropy permanent bounds}} \implies \boxed{\text{Hall-core bounds}},$$

$$\boxed{\text{Hall-core bounds}} \implies \boxed{\text{boundary exclusion}}.$$

The interior branch is

$$\boxed{\text{pairwise flatness and averaging convergence}} \implies \boxed{\text{positive maximizers converge to } U_n},$$

with strict local maximality providing the backward-uniqueness step. The main theorem combines these two branches only in Section 10.

2 Notation and the main theorem

Let S_n be the symmetric group. For $A = (a_{ij}) \in \mathbb{R}_{\geq 0}^{n \times n}$,

$$\text{per}(A) = \sum_{\sigma \in S_n} \prod_{i=1}^n a_{i, \sigma(i)}.$$

Let

$$K_n = \left\{ A \in \mathbb{R}_{\geq 0}^{n \times n} : \sum_{i,j=1}^n a_{ij} = n \right\}.$$

For $A \in K_n$, put

$$r_i = \sum_{j=1}^n a_{ij}, \quad c_j = \sum_{i=1}^n a_{ij},$$

and

$$\Phi(A) = \prod_{i=1}^n r_i + \prod_{j=1}^n c_j - \text{per}(A).$$

Let $J_n = \mathbf{1}\mathbf{1}^\top$, let

$$U_n = \frac{1}{n} J_n,$$

and define

$$\gamma_n = \text{per}(U_n) = \frac{n!}{n^n}.$$

A matrix is called *positive* if every entry is strictly positive. A matrix is *doubly stochastic* if every row sum and every column sum equals one. Symbols introduced inside a dimension-specific subsection are local to that subsection; repeated letters such as H , K , and q are always redefined before reuse. The global notation above remains fixed throughout.

Theorem 2.1 (Main theorem). *For every integer n with $6 \leq n \leq 15$ and every $A \in K_n$,*

$$\Phi(A) \leq 2 - \frac{n!}{n^n}.$$

Equality holds if and only if $A = U_n$.

The simplex K_n is compact and Φ is continuous, so global maximizers exist. We first develop two pieces of infrastructure that are valid in arbitrary dimension: permanent bounds from capacity and the classification of positive maximizers.

3 Stable-polynomial capacity and permanent bounds

3.1 Stability and capacity

Let

$$\mathbb{H} = \{z \in \mathbb{C} : \text{Im } z > 0\}.$$

A polynomial $p \in \mathbb{R}[x_1, \dots, x_m]$ is *real stable* if

$$p(z_1, \dots, z_m) \neq 0 \quad (z_1, \dots, z_m \in \mathbb{H}).$$

For a homogeneous polynomial p of degree m in m variables with nonnegative coefficients, define

$$\text{Cap}(p) = \inf_{x_1, \dots, x_m > 0} \frac{p(x_1, \dots, x_m)}{x_1 \cdots x_m}.$$

We also set $\text{Cap}(0) = 0$. If a nonzero polynomial in this quotient is independent of some denominator variable x_j , then its capacity is zero: fix the remaining variables and let $x_j \rightarrow \infty$. We shall use this observation to verify explicitly that every variable extracted below has positive degree. Set

$$G(1) = 1, \quad G(k) = \left(\frac{k-1}{k}\right)^{k-1} \quad (k \geq 2),$$

and

$$v_m = \frac{m!}{m^m} \quad (m \geq 1), \quad v_0 = 1.$$

The telescoping identity

$$\prod_{k=1}^m G(k) = v_m \tag{3.1}$$

will be used repeatedly. The sequence $G(k)$ is decreasing. Indeed, $G(k)^{-1} = (1 + 1/(k-1))^{k-1}$, and the function

$$x \mapsto x \log(1 + 1/x)$$

is increasing on $(0, \infty)$ because $\log(1 + y) > y/(1 + y)$ for $y > 0$.

We use two classical closure properties of real stability. They are included for completeness.

Lemma 3.1 (Closure under specialization and differentiation). *Let p be real stable.*

- (i) *Specializing one variable to a real number produces either a real-stable polynomial or the zero polynomial.*
- (ii) *Every nonzero partial derivative of p is real stable.*

Proof. For real specialization, fix $a \in \mathbb{R}$ and replace it first by $a + i\varepsilon$. The resulting polynomials are zero-free on the product of upper half-planes and converge locally uniformly as $\varepsilon \downarrow 0$. Hurwitz's theorem implies that the limit is either zero-free there or identically zero.

For differentiation, write $p(z', w) = \sum_{r=0}^d a_r(z') w^r$ with a_d not identically zero. The functions $(iR)^{-d} p(z', iR)$ are zero-free for $z' \in \mathbb{H}^{m-1}$ and converge locally uniformly to $a_d(z')$ as $R \rightarrow \infty$. Hurwitz's theorem shows that a_d is real stable, so $a_d(z') \neq 0$ for every $z' \in \mathbb{H}^{m-1}$. Thus, after fixing such a z' , the univariate polynomial $w \mapsto p(z', w)$ has exact degree d , and every one of its roots lies in the closed lower half-plane. By the Gauss–Lucas theorem, the roots of its derivative lie in the convex hull of those roots and hence also in the closed lower half-plane. Therefore the derivative does not vanish for $w \in \mathbb{H}$. Since z' was arbitrary, the partial derivative is real stable. The zero-polynomial case is excluded by hypothesis. $\square$

The next one-variable estimate is the quantitative core of the capacity method.

Lemma 3.2 (One-variable capacity estimate). *Let f be a nonzero univariate polynomial with nonnegative coefficients. Suppose all zeros of f are real and nonpositive, and let $d = \deg f \geq 1$. Then*

$$f'(0) \geq G(d) \inf_{t>0} \frac{f(t)}{t}. \quad (3.2)$$

Proof. Assume first that $f(0) > 0$. Then

$$f(t) = f(0) \prod_{\ell=1}^d (1 + \lambda_\ell t), \quad \lambda_\ell \geq 0.$$

Let

$$s = \sum_{\ell=1}^d \lambda_\ell = \frac{f'(0)}{f(0)}.$$

Since $d \geq 1$ is the exact degree, $s > 0$. By AM–GM,

$$\prod_{\ell=1}^d (1 + \lambda_\ell t) \leq \left(1 + \frac{st}{d}\right)^d.$$

If $d \geq 2$, choose $t = d/[s(d-1)]$. This gives

$$\inf_{t>0} \frac{f(t)}{t} \leq \frac{f'(0)}{G(d)}.$$

For $d = 1$, the same conclusion follows by letting $t \rightarrow \infty$.

If $f(0) = 0$, write $f(t) = t^r g(t)$ with $g(0) > 0$. For $r = 1$,

$$f'(0) = g(0) \geq G(d) \inf_{t>0} \frac{f(t)}{t},$$

because the infimum is at most the limit $g(0)$ as $t \downarrow 0$. If $r \geq 2$, then both $f'(0)$ and the infimum in (3.2) are zero. $\square$

Theorem 3.3 (Derivative-capacity inequality). *Let p be a homogeneous real-stable polynomial of degree m in m variables, with nonnegative coefficients. Put*

$$d = \deg_{x_m} p$$

and

$$q(x_1, \dots, x_{m-1}) = \left. \frac{\partial p}{\partial x_m} \right|_{x_m=0}.$$

If $d \geq 1$, then

$$\text{Cap}(q) \geq G(d) \text{Cap}(p). \quad (3.3)$$

Proof. If $q = 0$, then for every $x' > 0$ the polynomial $f(t) = p(x', t)$ satisfies $f'(0) = 0$. Lemma 3.2 therefore gives $\inf_{t>0} f(t)/t = 0$ for every x' , and hence $\text{Cap}(p) = 0$; the assertion follows from the convention $\text{Cap}(0) = 0$. We may consequently assume $q \neq 0$.

Fix $x' = (x_1, \dots, x_{m-1}) > 0$ and set

$$f(t) = p(x', t).$$

By Lemma 3.1, f is stable as a univariate real polynomial. Hence all its zeros are real: a nonreal zero below the real axis would have a conjugate zero above it. Nonnegative coefficients rule out positive zeros. Thus Lemma 3.2 applies.

For every $t > 0$, the definition of capacity gives

$$\frac{f(t)}{t} \geq \text{Cap}(p) x_1 \cdots x_{m-1}.$$

Therefore

$$q(x') = f'(0) \geq G(d) \text{Cap}(p) x_1 \cdots x_{m-1}.$$

Divide by $x_1 \cdots x_{m-1}$ and take the infimum over $x' > 0$. □

Corollary 3.4 (Square-free coefficient extraction). *Let p be homogeneous, real stable, of degree m in m variables, and have nonnegative coefficients. Then the coefficient of $x_1 \cdots x_m$ is at least*

$$v_m \text{Cap}(p).$$

Proof. If $\text{Cap}(p) = 0$, the assertion is immediate. Otherwise differentiate successively in $x_m, x_{m-1}, \dots, x_1$, setting each differentiated variable to zero. Theorem 3.3 keeps every intermediate capacity positive. At the stage with j variables remaining, the relevant degree is at most j and at least one; if it were zero, the polynomial would be independent of a denominator variable and would have capacity zero. Since G is decreasing, Theorem 3.3 contributes at least $G(j)$. Equation (3.1) completes the proof. □

3.2 The support-sensitive permanent inequality

Theorem 3.5 (Support-sensitive permanent bound). *Let $B = (b_{ij})$ be an $n \times n$ doubly stochastic matrix. Order its columns arbitrarily, and let d_j be the number of positive entries in the j th ordered column. Then*

$$\text{per}(B) \geq \prod_{j=1}^n G(\min\{j, d_j\}). \quad (3.4)$$

Proof. Consider

$$P_B(x_1, \dots, x_n) = \prod_{i=1}^n \left(\sum_{j=1}^n b_{ij} x_j \right).$$

Every linear factor has positive imaginary part when all variables lie in $\mathbb{H}$, so P_B is real stable. Weighted AM–GM gives, for $x_j > 0$,

$$\sum_j b_{ij} x_j \geq \prod_j x_j^{b_{ij}}.$$

Multiplying over the rows and using the column sums,

$$P_B(x) \geq x_1 \cdots x_n.$$

Equality holds at $x_1 = \cdots = x_n = 1$, and therefore

$$\text{Cap}(P_B) = 1. \quad (3.5)$$

Set $p_n = P_B$ and recursively

$$p_{j-1} = \frac{\partial p_j}{\partial x_j} \Big|_{x_j=0} \quad (j = n, n-1, \dots, 1).$$

At stage j , the polynomial has degree j in j variables. Differentiation in other variables cannot increase its degree in x_j , so

$$\deg_{x_j} p_j \leq \min\{j, d_j\}.$$

Every intermediate capacity is positive. Therefore p_j cannot be independent of x_j , for otherwise the denominator-variable observation above would give $\text{Cap}(p_j) = 0$. Thus $1 \leq \deg_{x_j} p_j \leq \min\{j, d_j\}$, and all hypotheses of Theorem 3.3 are met. Using the monotonicity of G and (3.5),

$$p_0 \geq \prod_{j=1}^n G(\min\{j, d_j\}).$$

The scalar p_0 is the coefficient of $x_1 \cdots x_n$ in P_B , namely $\text{per}(B)$. □

3.3 Zero-pattern and entropy consequences

Corollary 3.6 (One zero). *If a doubly stochastic B has at least one zero entry, then*

$$\text{per}(B) \geq \kappa_n, \quad \kappa_n := v_{n-1} G(n-1) = \frac{(n-2)!(n-2)^{n-2}}{(n-1)^{2n-4}}. \quad (3.6)$$

Moreover $\kappa_n > \gamma_n$ for every $n \geq 3$.

Proof. Put a column containing a zero last in Theorem 3.5. Its support size is at most $n-1$, giving (3.6).

For $m = n-1 \geq 2$,

$$\frac{\kappa_n}{\gamma_n} = \left(1 + \frac{1}{m}\right)^m \left(1 - \frac{1}{m}\right)^{m-1} = \left(1 + \frac{1}{m}\right) \left(1 - \frac{1}{m^2}\right)^{m-1}.$$

Bernoulli's inequality yields

$$\left(1 - \frac{1}{m^2}\right)^{m-1} \geq 1 - \frac{m-1}{m^2} = 1 - \frac{1}{m} + \frac{1}{m^2}.$$

Multiplication by $1 + 1/m$ gives $1 + 1/m^3 > 1$. □

Theorem 3.7 (Best-possible rectangular-zero-block bound). *Let B be doubly stochastic and suppose that, after row and column permutations, it contains an $a \times b$ zero block, where*

$$a, b \geq 1, \quad p := n - a - b \geq 1.$$

Then

$$\text{per}(B) \geq L_{n,a,b}^\# := \frac{v_{n-a}v_{n-b}}{v_p}. \quad (3.7)$$

The constant is best possible; equality is attained by the block-constant matrix displayed in the proof. No classification of all equality cases is asserted here.

Proof. Let C be the b columns in the zero block and let R be its a rows. In P_B , every variable indexed by C occurs only in the $n - a$ row factors indexed by R^c .

Differentiate successively in the b variables indexed by C , setting each differentiated variable to zero. After k such differentiations, every surviving product-rule term has differentiated k distinct row factors: each row factor is linear and cannot be differentiated twice. Consequently, the degree of the next special variable is at most $n - a - k$. The current capacity remains positive, so that degree is also at least one; otherwise the current polynomial would omit a denominator variable and have zero capacity. Theorem 3.3 therefore supplies the factor

$$\prod_{k=0}^{b-1} G(n - a - k) = \prod_{j=p+1}^{n-a} G(j) = \frac{v_{n-a}}{v_p}.$$

A homogeneous stable polynomial of degree $n - b$ in $n - b$ variables remains. Corollary 3.4 supplies the additional factor v_{n-b} . Since $\text{Cap}(P_B) = 1$, (3.7) follows.

To prove best-possible attainment, consider

$$B_0 = \begin{pmatrix} 0_{a \times b} & \frac{1}{n-b} J_{a,n-b} \\ \frac{1}{n-a} J_{n-a,b} & \frac{p}{(n-a)(n-b)} J_{n-a,n-b} \end{pmatrix}.$$

It is doubly stochastic. Every contributing permutation uses a upper-right entries, b lower-left entries, and p lower-right entries. Their number is

$$\frac{(n-b)!}{p!} \cdot \frac{(n-a)!}{p!} \cdot p! = \frac{(n-a)!(n-b)!}{p!}.$$

Multiplying by the common weight gives

$$\text{per}(B_0) = \frac{(n-a)!(n-b)!}{p!} \frac{p^p}{(n-a)^{n-a}(n-b)^{n-b}} = \frac{v_{n-a}v_{n-b}}{v_p}.$$

□

For the middle dimensions it is convenient to record a weaker consequence with a simpler form.

Corollary 3.8. *Suppose $1 \leq a \leq b$ and a doubly stochastic B contains an $a \times b$ zero block. Then*

$$\text{per}(B) \geq L_{n,a,b} := v_{n-a}G(n-b)^a. \quad (3.8)$$

In particular, a 2×2 zero block gives

$$\text{per}(B) \geq v_{n-2}G(n-2)^2. \quad (3.9)$$

Proof. Transpose first. The transposed zero block has a columns, each supported on at most $n - b$ rows. Put those a columns last in Theorem 3.5; the first $n - a$ factors give v_{n-a} and the last a factors give $G(n - b)^a$. $\square$

Theorem 3.9 (Column-entropy permanent bound). *Let B be doubly stochastic, and let $b = (b_1, \dots, b_n)$ be any one of its columns. Put*

$$s(b) = \sum_{i=1}^n \left(b_i - \frac{1}{n}\right)^2.$$

Then

$$\text{per}(B) \geq \gamma_n \exp\left(\frac{s(b)}{2}\right). \quad (3.10)$$

Proof. It is enough first to consider $0 < b_i < 1$. For a general doubly stochastic matrix, apply the positive-case argument to $B_\varepsilon = (1 - \varepsilon)B + \varepsilon U_n$. The entropy expression is continuous under $0 \log 0 = 0$, and the permanent is polynomial in the entries, so the asserted inequality follows by letting $\varepsilon \downarrow 0$. Take the selected column to be the last one and set

$$L_i(x_1, \dots, x_{n-1}) = \sum_{j < n} b_{ij} x_j.$$

The derivative polynomial is

$$q = \frac{\partial P_B}{\partial x_n} \Big|_{x_n=0} = \sum_i b_i \prod_{k \neq i} L_k.$$

Weighted AM–GM, with weights b_i , gives

$$q \geq \prod_i L_i^{1-b_i}.$$

A second weighted AM–GM inequality gives

$$L_i \geq (1 - b_i) \prod_{j < n} x_j^{b_{ij}/(1-b_i)}.$$

Therefore

$$q(x) \geq \left(\prod_i (1 - b_i)^{1-b_i} \right) x_1 \cdots x_{n-1},$$

so

$$\text{Cap}(q) \geq \prod_i (1 - b_i)^{1-b_i}.$$

Corollary 3.4 yields

$$\text{per}(B) \geq v_{n-1} \prod_i (1 - b_i)^{1-b_i}. \quad (3.11)$$

Let $h(t) = (1 - t) \log(1 - t)$. Since $h''(t) = 1/(1 - t) \geq 1$ on $[0, 1]$,

$$h(t) \geq h(1/n) + h'(1/n)(t - 1/n) + \frac{1}{2}(t - 1/n)^2.$$

Summing at $t = b_i$, the linear terms cancel. Hence

$$\sum_i h(b_i) \geq nh(1/n) + \frac{1}{2}s(b).$$

Now $\exp(nh(1/n)) = G(n)$ and $v_{n-1}G(n) = \gamma_n$. Substitution into (3.11) proves (3.10). $\square$

3.4 Complement entropy and core-marginal refinements

The column-entropy estimate retains one column of a doubly stochastic matrix. The lower-dimensional boundary arguments also require the dual information carried by all complementary minors of a column and, in Dimension 6, by the marginal distributions of a whole Hall core.

For a probability vector $q = (q_1, \dots, q_m)$, put

$$\Psi_m(q) = \sum_{i=1}^m (1 - q_i) \log(1 - q_i) - m \left(1 - \frac{1}{m}\right) \log \left(1 - \frac{1}{m}\right), \quad (3.12)$$

with the convention $0 \log 0 = 0$, and write

$$s_m(q) = \sum_{i=1}^m \left(q_i - \frac{1}{m}\right)^2.$$

Lemma 3.10 (Quadratic complement-entropy bounds). *For every probability vector q of length $m \geq 2$,*

$$\Psi_m(q) \geq \left(\frac{1}{2} + \frac{1}{3m}\right) s_m(q) \geq \frac{1}{2} s_m(q). \quad (3.13)$$

Proof. Let $h(x) = (1 - x) \log(1 - x)$. Since

$$h''(x) = \frac{1}{1 - x} \geq 1 + x \quad (0 \leq x < 1),$$

For $q_i < 1$, Taylor's formula with integral remainder about $1/m$ gives, for $d_i = q_i - 1/m$,

$$h(q_i) \geq h(1/m) + h'(1/m)d_i + \frac{1 + 1/m}{2} d_i^2 + \frac{1}{6} d_i^3.$$

Because $d_i \geq -1/m$, one has $d_i^3 \geq -m^{-1}d_i^2$. Summation and $\sum_i d_i = 0$ prove the first inequality. Boundary coordinates $q_i = 1$ follow by continuity. The second inequality is immediate. $\square$

Theorem 3.11 (Complementary-minor entropy). *Let M be an $m \times m$ doubly stochastic matrix, let $q = (q_1, \dots, q_m)$ be one of its columns, and let*

$$d_i = \text{per } M(i \mid j_0)$$

be the permanent of the submatrix obtained by deleting row i and the selected column j_0 . With the convention that a factor of exponent zero is omitted,

$$\prod_{i=1}^m d_i^{q_i} \geq v_m \exp(\Psi_m(q)). \quad (3.14)$$

Consequently,

$$\prod_{i=1}^m d_i^{q_i} \geq v_m \exp \left\{ \left(\frac{1}{2} + \frac{1}{3m} \right) s_m(q) \right\} \geq v_m \exp \left(\frac{s_m(q)}{2} \right). \quad (3.15)$$

Proof. It is enough first to treat the positive case. For a general M , apply the argument to $M_\varepsilon = (1 - \varepsilon)M + \varepsilon U_m$ and let $\varepsilon \downarrow 0$. All selected-column coordinates and complementary minors are positive for $\varepsilon > 0$. If $q_i(0) = 0$, then $q_i(\varepsilon) = O(\varepsilon)$, while $d_i(\varepsilon)$ is a nonzero polynomial in ε ; hence $q_i(\varepsilon) \log d_i(\varepsilon) = O(\varepsilon |\log \varepsilon|)$. If $q_i(0) > 0$, then $d_i(0) > 0$: otherwise the left side of the positive- ε inequality would tend to zero, whereas its right side tends to

the strictly positive number $v_m \exp(\Psi_m(q(0)))$. Thus every logarithmic term has the asserted limit, and the product inequality passes to $\varepsilon = 0$.

Take the selected column to be the last one and write

$$L_i(x_1, \dots, x_{m-1}) = \sum_{j < m} m_{ij} x_j.$$

For arbitrary $t_1, \dots, t_m > 0$, set

$$q_t(x) = \sum_{i=1}^m q_i t_i \prod_{k \neq i} L_k(x).$$

This is homogeneous, real stable, and of degree $m - 1$, because

$$q_t = \frac{\partial}{\partial z} \prod_{i=1}^m (L_i(x) + q_i t_i z) \Big|_{z=0}.$$

Weighted AM–GM, first with weights q_i and then within each row, gives

$$\begin{aligned} q_t &= \left(\prod_i L_i \right) \sum_i q_i \frac{t_i}{L_i} \\ &\geq \left(\prod_i t_i^{q_i} \right) \prod_i L_i^{1-q_i} \\ &\geq \left(\prod_i t_i^{q_i} \right) \left(\prod_i (1 - q_i)^{1-q_i} \right) x_1 \cdots x_{m-1}. \end{aligned}$$

Thus

$$\text{Cap}(q_t) \geq \prod_i t_i^{q_i} \prod_i (1 - q_i)^{1-q_i}.$$

The coefficient of $x_1 \cdots x_{m-1}$ in q_t is $\sum_i q_i t_i d_i$, so Corollary 3.4 yields

$$\sum_i q_i t_i d_i \geq v_{m-1} \prod_i t_i^{q_i} \prod_i (1 - q_i)^{1-q_i}.$$

This inequality implies $d_i > 0$ whenever $q_i > 0$. Put $t_i = d_i^{-1}$ on the support of q . Since $\sum_i q_i = 1$,

$$\prod_i d_i^{q_i} \geq v_{m-1} \prod_i (1 - q_i)^{1-q_i} = v_m \exp(\Psi_m(q)),$$

where $v_{m-1} G(m) = v_m$. Lemma 3.10 gives the stated consequences. $\square$

The case in which a critical zero block leaves only one unit of mass in the opposite corner will be called a *single-bridge* configuration.

Theorem 3.12 (Single-bridge zero-block entropy). *Let B be an $n \times n$ doubly stochastic matrix with a zero block $B[R, C] = 0$, where $|R| = a$, $|C| = b$, and $a + b = n - 1$. Put $I = R^c$, $J = C^c$, $m = b + 1$, and $\ell = a + 1$. The core $X = B[I, J]$ has total mass one. Let ξ and η be its row- and column-sum vectors, and put*

$$s_I = s_m(\xi), \quad s_J = s_\ell(\eta).$$

Then

$$\text{per}(B) \geq v_m v_\ell \exp(\Psi_m(\xi) + \Psi_\ell(\eta)). \quad (3.16)$$

In particular, with $c_q = \frac{1}{2} + \frac{1}{3q}$,

$$\text{per}(B) \geq v_m v_\ell \exp(c_m s_I + c_\ell s_J) \geq v_m v_\ell \exp\left(\frac{s_I + s_J}{2}\right). \quad (3.17)$$

The constant $v_m v_\ell$ is $L_{n,a,b}^\sharp$.

Proof. Every perfect matching uses exactly one edge of X . For $i \in I$ and $j \in J$, put

$$e_i = \text{per } B[I \setminus \{i\}, C], \quad d_j = \text{per } B[R, J \setminus \{j\}].$$

Expansion according to the unique core edge gives

$$\text{per}(B) = \sum_{i \in I, j \in J} x_{ij} e_i d_j.$$

The x_{ij} sum to one, so weighted AM–GM implies

$$\text{per}(B) \geq \prod_{i \in I} e_i^{\xi_i} \prod_{j \in J} d_j^{\eta_j}.$$

Adjoin ξ as a last column to $B[I, C]$. The resulting $m \times m$ matrix is doubly stochastic and its complementary minors are the e_i . Similarly, adjoining η to the transpose of $B[R, J]$ gives an $\ell \times \ell$ doubly stochastic matrix whose complementary minors are the d_j . Apply Theorem 3.11 on both sides, and then Lemma 3.10. $\square$

The next estimate retains one supported column inside an arbitrary rectangular zero block.

Theorem 3.13 (Distinguished zero-block column). *Let B be an $n \times n$ doubly stochastic matrix containing an $a \times b$ zero block $B[R, C] = 0$, where*

$$a, b \geq 1, \quad p = n - a - b \geq 1.$$

Put $I = R^c$ and $m = n - a$. Select a column $j_0 \in C$, and let $q = (b_{ij_0})_{i \in I}$ be its probability vector on I . Then

$$\text{per}(B) \geq L_{n,a,b}^\# \exp(\Psi_m(q)). \quad (3.18)$$

Consequently,

$$\text{per}(B) \geq L_{n,a,b}^\# \exp\left(\frac{s_m(q)}{2}\right).$$

The transposed assertion holds for a selected row in R .

Proof. First assume that B is positive on every position outside the prescribed zero block. Differentiate the product polynomial P_B first in the selected column variable and set that variable to zero. With

$$L_i(x) = \sum_{j \neq j_0} b_{ij} x_j,$$

the resulting homogeneous real-stable polynomial is

$$Q(x) = \sum_{i \in I} q_i \prod_{k \neq i} L_k(x).$$

Exactly as in the column-entropy proof, two weighted AM–GM inequalities give

$$\text{Cap}(Q) \geq \prod_{i \in I} (1 - q_i)^{1 - q_i}.$$

Differentiate next in the remaining $b - 1$ variables indexed by C . Their successive degrees are at most

$$m - 1, m - 2, \dots, p + 1.$$

At every stage the current capacity is positive, so the variable being extracted has degree at least one. Thus the derivative-capacity theorem is applicable at each step and supplies

v_{m-1}/v_p . Square-free coefficient extraction in the remaining $n - b$ variables supplies v_{n-b} . Therefore

$$\text{per}(B) \geq \frac{v_{m-1}v_{n-b}}{v_p} \prod_{i \in I} (1 - q_i)^{1-q_i}.$$

Since $v_{m-1}G(m) = v_m$, this is exactly (3.18).

For a general B , mix it with the block-constant attaining matrix from Theorem 3.7. This preserves the prescribed zero block and is positive on every allowed entry. Let the mixing parameter tend to zero; the permanent is polynomial in the entries and the entropy term is continuous under the convention $0 \log 0 = 0$. $\square$

We shall use the distinguished-column estimate through the following symmetric consequence. Let $B[R, C] = 0$ be as above, put

$$I = R^c, \quad J = C^c, \quad m = n - a, \quad \ell = n - b,$$

and note that the core $B[I, J]$ has total mass p . Define its normalized row and column marginals by

$$\xi_i = \frac{1}{p} \sum_{j \in J} b_{ij}, \quad \eta_j = \frac{1}{p} \sum_{i \in I} b_{ij},$$

and set $S_\xi = s_m(\xi)$ and $S_\eta = s_\ell(\eta)$.

Proposition 3.14 (Core-marginal zero-block entropy). *Let $L = L_{n,a,b}^\sharp$. If $p \geq 2$, then*

$$\text{per}(B) \geq L(1 + c_{a,b}(S_\xi + S_\eta)), \quad c_{a,b} = \frac{p^2}{2(a^2 + b^2)}. \quad (3.19)$$

If $p = 1$, then

$$\text{per}(B) \geq L(1 + c_m S_\xi + c_\ell S_\eta), \quad c_q = \frac{1}{2} + \frac{1}{3q}. \quad (3.20)$$

Proof. Assume first that $p \geq 2$. For each of the b columns indexed by C , view its entries on I as a probability vector. Their average is

$$\bar{q}_i = \frac{1 - p\xi_i}{b},$$

and hence

$$\sum_{i \in I} \left(\bar{q}_i - \frac{1}{m} \right)^2 = \frac{p^2}{b^2} S_\xi.$$

Theorem 3.13, convexity of Ψ_m , and Lemma 3.10 give

$$\log \frac{\text{per}(B)}{L} \geq \max_{j \in C} \Psi_m(q^{(j)}) \geq \frac{1}{b} \sum_{j \in C} \Psi_m(q^{(j)}) \geq \Psi_m(\bar{q}) \geq \frac{p^2}{2b^2} S_\xi.$$

Transposition gives

$$\log \frac{\text{per}(B)}{L} \geq \frac{p^2}{2a^2} S_\eta.$$

The maximum of these two lower bounds is at least their weighted mean with weights $b^2/(a^2 + b^2)$ and $a^2/(a^2 + b^2)$. Thus

$$\log \frac{\text{per}(B)}{L} \geq \frac{p^2}{2(a^2 + b^2)} (S_\xi + S_\eta).$$

Use $\exp x \geq 1 + x$.

If $p = 1$, apply Theorem 3.12 and then $\exp x \geq 1 + x$. $\square$

4 Positive global maximizers are uniform

This section gives an independent proof of the interior classification. No permanent lower bound is used.

4.1 Flatness under pairwise averaging

Choose two columns u, v of a positive matrix $A \in K_n$. For $t \in \mathbb{R}$, replace them by

$$u(t) = (1 - t)u + tv, \quad v(t) = tu + (1 - t)v, \quad (4.1)$$

and leave every other column fixed. Denote the resulting matrix by $A(t)$.

Lemma 4.1 (Pairwise averaging is flat at a positive maximizer). *If A is a positive global maximizer of Φ , then*

$$\Phi(A(t)) = \Phi(A) \quad \text{for every } t \in \mathbb{R}. \quad (4.2)$$

In particular, replacing two columns by their arithmetic mean preserves the global maximum value. The same statement holds for two rows.

Proof. Because A is positive, $A(t) \in K_n$ for all t in an open interval containing zero. The row sums are unchanged along the path. The product of the column sums is a quadratic polynomial in t , and multilinearity of the permanent in the two changed columns shows that $\text{per}(A(t))$ is also quadratic. Thus

$$F(t) := \Phi(A(t))$$

has degree at most two.

Replacing t by $1 - t$ interchanges the two columns, so $F(t) = F(1 - t)$. Hence

$$F(t) = F(0) + \lambda t(1 - t)$$

for some real λ . Since $A = A(0)$ is a local maximum along a two-sided feasible interval, $F'(0) = 0$, and therefore $\lambda = 0$. The row version follows by transposition. $\square$

4.2 Strict local maximality of the uniform matrix

Lemma 4.2 (Hessian at the uniform matrix). *Let $X = (x_{ij})$ satisfy*

$$\sum_{i,j} x_{ij} = 0.$$

Put

$$u_i = \sum_j x_{ij}, \quad v_j = \sum_i x_{ij},$$

and

$$Q = \sum_{i,j} x_{ij}^2, \quad R_2 = \sum_i u_i^2, \quad C_2 = \sum_j v_j^2.$$

Then

$$\left. \frac{d^2}{dt^2} \Phi(U_n + tX) \right|_{t=0} = -(1 - c_n)(R_2 + C_2) - c_n Q, \quad (4.3)$$

where

$$c_n = \frac{(n-2)!}{n^{n-2}}.$$

Consequently, the Hessian of Φ at U_n is negative definite on the tangent hyperplane of K_n .

Proof. The row sums of $U_n + tX$ are $1 + tu_i$. Since $\sum_i u_i = 0$,

$$\left. \frac{d^2}{dt^2} \prod_i (1 + tu_i) \right|_{t=0} = -R_2.$$

The column-product contribution is similarly $-C_2$.

For the permanent, direct differentiation gives

$$\left. \frac{d^2}{dt^2} \text{per}(U_n + tX) \right|_{t=0} = c_n \sum_{\substack{i \neq k \\ j \neq \ell}} x_{ij} x_{k\ell}. \quad (4.4)$$

Indeed, for fixed ordered pairs $i \neq k$ and $j \neq \ell$, there are $(n-2)!$ permutations satisfying $\sigma(i) = j$ and $\sigma(k) = \ell$, while the remaining $n-2$ entries contribute $n^{-(n-2)}$.

Expanding the identity

$$0 = \left(\sum_{i,j} x_{ij} \right)^2$$

according to whether two entries have the same row, the same column, or neither, gives

$$\sum_{\substack{i \neq k \\ j \neq \ell}} x_{ij} x_{k\ell} = Q - R_2 - C_2. \quad (4.5)$$

Substituting (4.5) into (4.4) proves (4.3). Since $0 < c_n \leq 1$ and $Q > 0$ for $X \neq 0$, the right side is strictly negative. $\square$

Corollary 4.3. *The matrix U_n is a strict local maximizer of Φ on K_n .*

Proof. For a tangent vector $X = (x_{ij})$, put $u_i = \sum_j x_{ij}$ and $v_j = \sum_i x_{ij}$. Since $\sum_{i,j} x_{ij} = 0$, one has $\sum_i u_i = \sum_j v_j = 0$. Direct differentiation at U_n gives

$$D\Phi(U_n)[X] = \sum_i u_i + \sum_j v_j - \frac{(n-1)!}{n^{n-1}} \sum_{i,j} x_{ij} = 0.$$

The tangent space of the affine hull of K_n is finite-dimensional. By Lemma 4.2, the quadratic form given by the Hessian is uniformly negative on its unit sphere. The vanishing linear term and Taylor's formula then give a relative neighborhood $\mathcal{N}$ of U_n in K_n such that

$$A \in \mathcal{N}, A \neq U_n \implies \Phi(A) < \Phi(U_n).$$

$\square$

4.3 Averaging convergence and the interior classification

Lemma 4.4 (Convergence of cyclic pairwise averaging). *Let $c_1, \dots, c_n$ be vectors, and repeatedly replace pairs by their arithmetic mean according to a cyclic schedule containing every unordered pair. Then all vectors converge to their common initial mean.*

Proof. Let $\bar{c} = n^{-1} \sum_j c_j$, which is invariant, and define

$$V = \sum_j \|c_j - \bar{c}\|^2.$$

Averaging c_s, c_t decreases V by

$$\frac{1}{2} \|c_s - c_t\|^2. \quad (4.6)$$

Thus V decreases to a limit $L \geq 0$.

Assume $L > 0$. Take a convergent subsequence of the states at the beginnings of complete cycles. The total decrease during each selected cycle tends to zero. Since that decrease is a finite sum of the nonnegative quantities in (4.6), every pair difference at the moment it is averaged tends to zero. Passing to the limiting cycle, every averaging operation makes no change. Since every unordered pair occurs, all limiting vectors are equal, forcing $V = 0$, a contradiction. Hence $L = 0$. $\square$

Theorem 4.5 (Positive-maximizer theorem). *For every $n \geq 2$, every positive global maximizer of Φ on K_n is U_n .*

Proof. Let A be a positive global maximizer. Repeatedly average pairs of columns in a cyclic schedule containing every unordered pair. By Lemma 4.1, every matrix in the sequence is a positive global maximizer with the same value as A . By Lemma 4.4, the sequence converges to a positive matrix C with all columns equal. Continuity shows that C is also a global maximizer.

Now cyclically average pairs of rows of C . Every matrix remains a positive global maximizer, and the sequence converges to U_n : the equal columns of C each have sum one, and row averaging preserves every column sum. Therefore

$$\Phi(U_n) = \max_{K_n} \Phi. \quad (4.7)$$

Let Y_m be the row-averaging sequence converging to U_n . Corollary 4.3 implies that $Y_m = U_n$ for all sufficiently large m , because every Y_m has the value $\Phi(U_n)$.

We claim that one pair-averaging operation cannot map a distinct positive global maximizer X to U_n . If it did, the path (4.1) would have constant value by Lemma 4.1, and points with $t \neq 1/2$ arbitrarily close to $1/2$ would be distinct from U_n , arbitrarily close to it, and have value $\Phi(U_n)$. This contradicts strict local maximality. Hence the predecessor of U_n under such an averaging operation is also U_n .

Applying this observation backward through the finite row-averaging segment gives $C = U_n$. The column-averaging sequence converges to U_n and has constant value, so strict local maximality again forces some finite column-averaging iterate to equal U_n . Applying the same backward argument gives $A = U_n$. $\square$

Remark 4.6. Theorem 4.5 reproduces the positive-support conclusion historically due to Hwang, but its proof here is independent of [5].

5 Joint deficits, subset sums, and Hall dilation

Let A be a global maximizer of Φ on K_n . Since U_n is feasible,

$$\Phi(A) \geq \Phi(U_n) = 2 - \gamma_n. \quad (5.1)$$

Write

$$\prod_i r_i = 1 - \rho, \quad \prod_j c_j = 1 - \sigma, \quad \text{per}(A) = \gamma_n - \delta. \quad (5.2)$$

Lemma 5.1 (Joint deficit budget). *The quantities in (5.2) satisfy*

$$0 \leq \delta \leq \gamma_n, \quad \rho \geq 0, \quad \sigma \geq 0, \quad \rho + \sigma \leq \delta. \quad (5.3)$$

In particular, all row sums and column sums of A are positive.

Proof. The row sums and column sums each have arithmetic mean one, so AM–GM gives

$$\prod_i r_i \leq 1, \quad \prod_j c_j \leq 1.$$

Thus $\rho, \sigma \geq 0$. Also

$$\text{per}(A) = \prod_i r_i + \prod_j c_j - \Phi(A) \leq 2 - (2 - \gamma_n) = \gamma_n,$$

which gives $0 \leq \delta \leq \gamma_n$. Substitution into (5.1) yields

$$2 - \rho - \sigma - \gamma_n + \delta \geq 2 - \gamma_n,$$

and hence $\rho + \sigma \leq \delta$. If a row or column sum vanished, one of ρ, σ would equal one, contradicting $\delta \leq \gamma_n < 1$. $\square$

5.1 Hall domination

For a nonnegative matrix S and index sets $I, J \subseteq [n]$, write

$$s_S(I, J) = \sum_{i \in I, j \in J} s_{ij}.$$

Lemma 5.2 (Hall domination). *A nonnegative $n \times n$ matrix S dominates a doubly stochastic matrix entrywise if and only if*

$$s_S(I, J) \geq |I| + |J| - n \quad (5.4)$$

for every $I, J \subseteq [n]$ with $|I| + |J| > n$.

If equality holds in (5.4) for some I, J , then every doubly stochastic $B \leq S$ satisfies

$$B[I^c, J^c] = 0. \quad (5.5)$$

Proof. Construct a flow network with source-to-row capacities one, row-to-column capacities s_{ij} , and column-to-sink capacities one. A flow of value n is exactly a doubly stochastic matrix $B \leq S$. A cut having row set I and column set K on the source side has capacity

$$n - |I| + s_S(I, K^c) + |K|.$$

By max-flow/min-cut, every cut has capacity at least n exactly when (5.4) holds, after writing $J = K^c$.

Let

$$p = |I| + |J| - n > 0.$$

Every doubly stochastic B satisfies

$$s_B(I, J) - s_B(I^c, J^c) = p. \quad (5.6)$$

If $s_S(I, J) = p$ and $B \leq S$, then $s_B(I, J) \leq p$. Equation (5.6) and nonnegativity give the reverse inequality, so equality holds and $s_B(I^c, J^c) = 0$. $\square$

5.2 Relative-entropy subset estimates

For $0 < p, q < 1$, let

$$D(p||q) = p \log \frac{p}{q} + (1-p) \log \frac{1-p}{1-q}.$$

We shall use the integral identity

$$D(p||q) = \int_p^q \frac{u-p}{u(1-u)} du. \quad (5.7)$$

It follows by differentiating with respect to q . In every subset-excess application below one has $q \geq p$. On that oriented interval the integrand is nonnegative, and $u(1-u) \leq 1/4$ gives

$$D(p||q) \geq 4 \int_p^q (u-p) du = 2(q-p)^2.$$

The case $q < p$ follows by the same argument after reversing the limits (or by symmetry), so the binary Pinsker bound is

$$D(p||q) \geq 2(p-q)^2. \quad (5.8)$$

For $7 \leq n \leq 14$, define η_n by

n	7	8	9	10	11	12	13	14
η_n	1/47	1/80	1/135	1/230	1/350	1/650	1/1100	1/1800.

(5.9)

For $0 \leq k \leq n-1$, define

$$\alpha_{n,0} = 0$$

and, for $k \geq 1$,

$$\alpha_{n,k} = \begin{cases} 2n \left(\frac{k}{n} + \eta_n \right) \left(1 - \frac{k}{n} - \eta_n \right), & 2k < n, \\ \frac{2k(n-k)}{n}, & 2k \geq n. \end{cases} \quad (5.10)$$

The exact arithmetic in Appendix A verifies

$$\eta_n^2 > \frac{\gamma_n}{2n(1-\gamma_n)} \quad (5.11)$$

and

$$\frac{\lfloor (n-1)/2 \rfloor}{n} + \eta_n < \frac{1}{2} \quad (5.12)$$

for $7 \leq n \leq 14$.

Lemma 5.3 (Subset excess, dimensions 7 through 14). *Let $7 \leq n \leq 14$. If E is a set of k rows, $1 \leq k \leq n-1$, and*

$$\sum_{i \in E} r_i = k + \varepsilon, \quad \varepsilon > 0,$$

then

$$\varepsilon^2 \leq \frac{\alpha_{n,k}\rho}{1-\delta}. \quad (5.13)$$

The analogous estimate holds for column subsets, with σ in place of ρ .

Proof. AM–GM applied separately on E and E^c gives

$$1 - \rho \leq \left(1 + \frac{\varepsilon}{k} \right)^k \left(1 - \frac{\varepsilon}{n-k} \right)^{n-k}. \quad (5.14)$$

Put

$$p = \frac{k}{n}, \quad q = \frac{k + \varepsilon}{n}.$$

Taking logarithms in (5.14),

$$nD(p\|q) \leq -\log(1 - \rho) \leq \frac{\rho}{1 - \rho} \leq \frac{\rho}{1 - \delta}. \quad (5.15)$$

Pinsker's bound and Lemma 5.1 imply

$$(q - p)^2 \leq \frac{\gamma_n}{2n(1 - \gamma_n)} < \eta_n^2. \quad (5.16)$$

If $p < 1/2$, then (5.12) and (5.16) imply $q < p + \eta_n < 1/2$. Hence

$$u(1 - u) \leq (p + \eta_n)(1 - p - \eta_n) \quad (p \leq u \leq q).$$

Using (5.7),

$$nD(p\|q) \geq \frac{\varepsilon^2}{2n(p + \eta_n)(1 - p - \eta_n)} = \frac{\varepsilon^2}{\alpha_{n,k}}.$$

If $p \geq 1/2$, then $u(1 - u) \leq p(1 - p)$ for $u \in [p, q]$, giving the same conclusion with $\alpha_{n,k} = 2np(1 - p)$. Combine this lower bound with (5.15). $\square$

For $n = 15$, a sharper singleton constant is needed. Set

$$\alpha_{15,0} = 0, \quad \alpha_{15,1} = \frac{195}{98}, \quad \alpha_{15,k} = \frac{15}{2} \quad (2 \leq k \leq 14). \quad (5.17)$$

Lemma 5.4 (Subset excess in dimension 15). *The conclusion of Lemma 5.3 holds for $n = 15$ with the constants in (5.17).*

Proof. The bound $\alpha_{15,k} = 15/2$ follows from (5.8) exactly as in the preceding proof.

Suppose $k = 1$. The exact comparison $\gamma_{15} < 1/300000$, verified in Appendix A, gives

$$\varepsilon^2 \leq \frac{15\gamma_{15}}{2(1 - \gamma_{15})} < \frac{15}{2 \cdot 299999} < \frac{1}{196}.$$

Thus $0 < \varepsilon < 1/14$, and with $p = 1/15$ and $q = (1 + \varepsilon)/15$,

$$\frac{1}{15} < q < \frac{1}{14}.$$

On $[1/15, q]$,

$$u(1 - u) \leq \frac{13}{196}.$$

Equation (5.7) therefore gives

$$15D\left(\frac{1}{15}\|q\right) \geq \frac{98}{195}\varepsilon^2.$$

Combining this with (5.15) proves

$$\varepsilon^2 \leq \frac{195}{98} \frac{\rho}{1 - \delta}.$$

$\square$

5.3 The critical Hall deficit

Define

$$\tau = \max \left\{ 0, \max_{\substack{I, J \subseteq [n] \\ |I| + |J| > n}} \left(1 - \frac{s_A(I, J)}{|I| + |J| - n} \right) \right\}. \quad (5.18)$$

Assume $\tau > 0$, and choose a critical pair I, J . Put

$$a = |I^c|, \quad b = |J^c|, \quad p = n - a - b \geq 1. \quad (5.19)$$

Lemma 5.5 (Hall-deficit estimate). *With the constants $\alpha_{n,k}$ defined above,*

$$\tau^2 \leq c_{n,a,b} \frac{\delta}{1 - \delta}, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2}. \quad (5.20)$$

Whenever the right side is less than one, the matrix

$$S = \frac{A}{1 - \tau} \quad (5.21)$$

dominates a doubly stochastic matrix B , and every such B satisfies

$$B[I^c, J^c] = 0. \quad (5.22)$$

Every zero entry of A is also a zero entry of B .

Proof. Let

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c) \geq 0.$$

Inclusion–exclusion gives

$$s_A(I, J) = p - e - f + x.$$

Criticality therefore implies

$$\tau p = e + f - x \leq e_+ + f_+. \quad (5.23)$$

By the subset estimates,

$$e_+^2 \leq \frac{\alpha_{n,a}\rho}{1 - \delta}, \quad f_+^2 \leq \frac{\alpha_{n,b}\sigma}{1 - \delta}.$$

Cauchy–Schwarz and (5.3) give

$$(e_+ + f_+)^2 \leq \frac{(\alpha_{n,a} + \alpha_{n,b})(\rho + \sigma)}{1 - \delta} \leq \frac{(\alpha_{n,a} + \alpha_{n,b})\delta}{1 - \delta}.$$

Together with (5.23), this proves (5.20).

By the definition of τ , the matrix S satisfies every Hall inequality (5.4). Lemma 5.2 gives a doubly stochastic $B \leq S$. Equality holds for the critical pair, so (5.22) follows. Finally, S has every zero of A , and $B \leq S$. $\square$

If $\tau = 0$, Lemma 5.2 gives a doubly stochastic $B \leq A$. Since both matrices have total sum n , necessarily $B = A$. Thus a boundary maximizer with $\tau = 0$ would itself be doubly stochastic with a zero, and Corollary 3.6 would give

$$\text{per}(A) \geq \kappa_n > \gamma_n,$$

contrary to Lemma 5.1. Hence every boundary maximizer must have $\tau > 0$.

5.4 A common normalized block-variation identity

The stationarity identities used in Sections 6, 7, 8, and 9 are instances of the same calculation. Let $E \subseteq [n] \times [n]$ be a set of entries of total mass m_E , let

$$e_i = \sum_{j:(i,j) \in E} a_{ij}, \quad f_j = \sum_{i:(i,j) \in E} a_{ij},$$

and, for a permutation σ , let $k_E(\sigma) = |\{i : (i, \sigma(i)) \in E\}|$. Multiply the entries in E by t and then multiply the whole matrix by

$$\lambda(t) = \frac{n}{n + (t-1)m_E}$$

so that the total mass remains n . At a global maximizer, differentiation at $t = 1$ gives

$$0 = R \sum_i \frac{e_i}{r_i} + C \sum_j \frac{f_j}{c_j} - \sum_{\sigma \in S_n} k_E(\sigma) \prod_i a_{i, \sigma(i)} - m_E \Phi(A), \quad (5.24)$$

where $R = \prod_i r_i$ and $C = \prod_j c_j$. Indeed, the first three terms are the derivative before normalization, while homogeneity of degree n and $n\lambda'(1) = -m_E$ give the last term. Each later stationarity lemma specifies E , its row and column masses, and the matching count $k_E(\sigma)$; thus no normalization factor is implicit.

5.5 Scalar exclusion lemmas

Lemma 5.6 (First-order scalar criterion). *Suppose*

$$\text{per}(B) \geq L, \quad \tau^2 \leq c \frac{\delta}{1-\delta}, \quad 0 \leq \delta \leq \gamma_n, \quad 0 \leq \tau < 1.$$

Then

$$L(1-\tau)^n - (\gamma_n - \delta) \geq M_n^{(1)}(L, c), \quad (5.25)$$

where

$$M_n^{(1)}(L, c) = L - \gamma_n - \frac{n^2 c L^2}{4(1-\gamma_n)}. \quad (5.26)$$

In particular, $M_n^{(1)}(L, c) > 0$ contradicts

$$\text{per}(A) = \gamma_n - \delta \geq L(1-\tau)^n.$$

Proof. Bernoulli's inequality gives $(1-\tau)^n \geq 1 - n\tau$. Hence

$$\begin{aligned} L(1-\tau)^n - (\gamma_n - \delta) &\geq L - \gamma_n + \delta - nL\tau \\ &\geq L - \gamma_n + \delta - \frac{nL\sqrt{c}}{\sqrt{1-\gamma_n}}\sqrt{\delta}. \end{aligned}$$

Complete the square in $\sqrt{\delta}$. □

Lemma 5.7 (Second-order scalar criterion). *Suppose, in addition, that $0 \leq \tau \leq T$ and*

$$(1-t)^n \geq 1 - nt + \lambda t^2 \quad (0 \leq t \leq T). \quad (5.27)$$

Then

$$L(1-\tau)^n - (\gamma_n - \delta) \geq M_n^{(2)}(L, c), \quad (5.28)$$

where

$$M_n^{(2)}(L, c) = L - \gamma_n - \frac{n^2 L^2}{4(\lambda L + (1-\gamma_n)/c)}. \quad (5.29)$$

Proof. The Hall bound implies

$$\delta \geq \frac{1 - \gamma_n}{c} \tau^2.$$

Therefore

$$L(1 - \tau)^n - (\gamma_n - \delta) \geq L - \gamma_n - nL\tau + \left(\lambda L + \frac{1 - \gamma_n}{c} \right) \tau^2.$$

The minimum of the quadratic on the right is (5.29). $\square$

6 Boundary exclusion in dimensions eleven through fifteen

We now apply the common framework. Throughout this section, A denotes a hypothetical boundary global maximizer. As observed after Lemma 5.5, necessarily $\tau > 0$.

6.1 Dimensions eleven through fourteen: ordinary Hall pairs

For $11 \leq n \leq 14$, let

$$\mathcal{P}_n = \{(a, b) : 0 \leq a \leq b, a + b \leq n - 1\}.$$

Transposition permits the restriction $a \leq b$. For $(a, b) \in \mathcal{P}_n$, put

$$p = n - a - b, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2}, \quad (6.1)$$

and

$$\Lambda_{n,a,b} = \begin{cases} \kappa_n, & a = 0, \\ v_{n-a} G(n-b)^a, & a \geq 1. \end{cases} \quad (6.2)$$

If $a = 0$, the forced Hall block is empty, but B inherits a zero from A , so Corollary 3.6 gives the first line. If $a \geq 1$, Corollary 3.8 gives the second.

The constants in (5.10) satisfy $\alpha_{n,k} \leq n/2$: in the first branch this is $2nx(1-x) \leq n/2$, and in the second it is $2k(n-k)/n \leq n/2$. Hence $c_{n,a,b} \leq n$. Moreover

$$\frac{\gamma_{n+1}}{\gamma_n} = \left(\frac{n}{n+1} \right)^n < 1,$$

so $\gamma_n \leq \gamma_{11} < 1/1000$ for $11 \leq n \leq 14$. Therefore

$$\tau^2 \leq \frac{n\gamma_n}{1 - \gamma_n} < \frac{14}{999} < 1.$$

Thus the Hall dilation is valid.

The following proposition is a finite exact-rational calculation. Its formulas are explicit in (6.1), (6.2), and (5.26); Appendix B gives a complete standard-library implementation using rational arithmetic only.

Proposition 6.1 (Exact ordinary margins). *For the indicated sets of pairs,*

$$M_n^{(1)}(\Lambda_{n,a,b}, c_{n,a,b}) > 0.$$

More precisely, the minimum values are as follows:

n	minimizing pair	minimum (decimal for scale)
11	(0, 9)	1.919523×10^{-7}
12	(0, 11)	2.454520×10^{-8}
13	(0, 12)	3.656538×10^{-8}
14	(0, 13)	1.708143×10^{-8} .

(6.3)

The decimal values in this and later summary tables are only readability aids. The proof compares the exact fractions recorded by the checker and in `verification_manifest.json`. For $n = 11$, the pair $(0, 10)$ is excluded from this proposition and is handled separately below.

Proof. Substitute the rational definitions into (5.26) for every pair in $\mathcal{P}_n$, omitting $(0, 10)$ when $n = 11$. There are respectively 35, 42, 49, 56 cases. Clearing denominators gives positive integers in every case. The exact enumeration in Appendix B also compares the resulting fractions and returns the minima displayed in (6.3). No rounded quantity is used. $\square$

Corollary 6.2. *All critical Hall pairs in dimensions 12, 13, 14 are impossible. In dimension 11, every critical pair except $(a, b) = (0, 10)$ and its transpose is impossible.*

Proof. For a critical pair, Lemma 5.5 gives B and the bound (5.20); the appropriate permanent estimate is (6.2). Proposition 6.1 and Lemma 5.6 contradict (6.4), where

$$\text{per}(A) = (1 - \tau)^n \text{per}(S) \geq (1 - \tau)^n \text{per}(B). \quad (6.4)$$

$\square$

6.2 The exceptional deficient column in dimension eleven

Assume $n = 11$ and $(a, b) = (0, 10)$. Then $I = [11]$, while J consists of one column, say column j , and

$$c_j = 1 - \tau. \quad (6.5)$$

The complementary set of ten columns has total sum $10 + \tau$. Lemma 5.3 gives

$$\tau^2 \leq \frac{\alpha_{11,10}\sigma}{1 - \delta}. \quad (6.6)$$

We obtain a second use of τ from stationarity.

Lemma 6.3 (Deficient-column stationarity). *Let A be a global maximizer in dimension 11. Suppose its row sums are positive and a column has sum $1 - \tau < 1$. If*

$$d = \max_i (r_i - 1)_+,$$

then

$$d \geq (1 - \gamma_{11})\tau. \quad (6.7)$$

Proof. Apply the normalized block-variation identity (5.24) to the deficient column. Here the block mass is $m_E = 1 - \tau$, its row masses are a_{ij} , its only nonzero column mass is $1 - \tau$, and every permanent monomial uses exactly one entry of that column. With

$$u = \sum_i \frac{a_{ij}}{r_i}, \quad K = \frac{\Phi(A)}{\prod_i r_i},$$

identity (5.24) becomes

$$\left(\prod_i r_i \right) u + \prod_j c_j - \text{per}(A) - (1 - \tau)\Phi(A) = 0.$$

Since $\prod_j c_j - \text{per}(A) = \Phi(A) - \prod_i r_i$, this simplifies to

$$u = 1 - K\tau. \quad (6.8)$$

Now put $w_i = a_{ij}/r_i$. Then $w_i \geq 0$, $\sum_i w_i = u$, and

$$(1 - \tau) - u = \sum_i w_i(r_i - 1) = (K - 1)\tau.$$

The right side is positive. Moreover $0 < u < 1$ by (6.8), so

$$(K - 1)\tau \leq d \sum_i w_i = du \leq d.$$

Finally, $\Phi(A) \geq 2 - \gamma_{11}$ and $\prod_i r_i \leq 1$, so $K \geq 2 - \gamma_{11}$. $\square$

Applying Lemma 5.3 to a singleton row attaining d and using Lemma 6.3,

$$\rho \geq (1 - \delta) \frac{(1 - \gamma_{11})^2}{\alpha_{11,1}} \tau^2. \quad (6.9)$$

Equation (6.6) gives

$$\sigma \geq (1 - \delta) \frac{1}{\alpha_{11,10}} \tau^2. \quad (6.10)$$

Together with $\rho + \sigma \leq \delta$,

$$\tau^2 \leq c_* \frac{\delta}{1 - \delta}, \quad (6.11)$$

where

$$c_* = \left(\frac{(1 - \gamma_{11})^2}{\alpha_{11,1}} + \frac{1}{\alpha_{11,10}} \right)^{-1} = \frac{1540632960668022501507138780}{1671236290061169784540151329}. \quad (6.12)$$

Exact subtraction gives

$$\frac{93}{100} - c_* = \frac{1361678908886539811520195597}{167123629006116978454015132900} > 0. \quad (6.13)$$

Since B inherits a zero, $\text{per}(B) \geq \kappa_{11}$. For fixed $L > 0$, formula (5.26) shows that $M_n^{(1)}(L, c)$ is strictly decreasing in c . Hence the certified upper bound $c_* < 93/100$ permits the larger, rationally convenient value $93/100$. Finally,

$$M_{11}^{(1)}\left(\kappa_{11}, \frac{93}{100}\right) > 0, \quad M_{11}^{(1)}\left(\kappa_{11}, \frac{93}{100}\right) \approx 1.250887 \times 10^{-7}. \quad (6.14)$$

The exact numerator and denominator are recorded literally in the machine-readable verification manifest and checked by the standard-library verifier. Lemma 5.6 gives a contradiction. Therefore dimension 11 has no boundary global maximizer.

6.3 Dimension fifteen: a thin–thick dichotomy

Let $n = 15$, and retain the notation (a, b, p) from (5.19). Call a critical pair *thin* if

$$a = 0, \quad b = 0, \quad \text{or} \quad \min\{a, b\} = 1.$$

Otherwise call it *thick*. Since $\alpha_{15,k} \leq 15/2$ for every k , Lemma 5.5 and (6.19) give

$$\tau^2 \leq \frac{15\gamma_{15}}{1 - \gamma_{15}} < \frac{15}{299999} < 1.$$

Thus the Hall dilation and the doubly stochastic matrix B are available in both cases.

For a thin pair, (5.17) gives

$$\alpha_{15,a} + \alpha_{15,b} \leq \frac{195}{98} + \frac{15}{2} = \frac{465}{49}.$$

Since $p \geq 1$,

$$\tau^2 \leq \frac{465}{49} \frac{\delta}{1-\delta}. \quad (6.15)$$

The matrix B inherits a zero, so

$$\text{per}(B) \geq \mu_1, \quad \mu_1 := v_{14}G(14) = \frac{13! 13^{13}}{14^{26}}. \quad (6.16)$$

For a thick pair, $a, b \geq 2$, so B contains a 2×2 zero block. Thus

$$\text{per}(B) \geq \mu_2, \quad \mu_2 := v_{13}G(13)^2 = \frac{13! 12^{24}}{13^{37}}. \quad (6.17)$$

Also

$$\tau^2 \leq 15 \frac{\delta}{1-\delta}. \quad (6.18)$$

The exact comparisons

$$\gamma_{15} < \frac{1}{300000}, \quad \gamma_{15} < \frac{29863}{10^{10}}, \quad (6.19)$$

$$\frac{29937}{10^{10}} < \mu_1 < \frac{3}{10^6}, \quad (6.20)$$

and

$$\frac{301}{10^8} < \mu_2 < \frac{4}{10^6} \quad (6.21)$$

are certified by the six positive integers listed in Appendix A.

For the thin case,

$$\frac{225}{4} \frac{465}{49} = \frac{104625}{196} < 534.$$

Equations (6.19) and (6.20) give

$$\mu_1 - \gamma_{15} > \frac{74}{10^{10}}$$

and

$$\frac{225(465/49)\mu_1^2}{4(1-\gamma_{15})} < 534 \cdot \frac{9}{10^{12}} \cdot \frac{300000}{299999} < \frac{5}{10^9}.$$

The final strict inequality follows after clearing denominators; its integer gap is

$$58\,195\,000\,000\,000\,000 > 0.$$

Therefore

$$M_{15}^{(1)} \left(\mu_1, \frac{465}{49} \right) > \frac{74}{10^{10}} - \frac{5}{10^9} = \frac{24}{10^{10}} > 0. \quad (6.22)$$

For the thick case, $\gamma_{15} < 299/10^8$, so

$$\mu_2 - \gamma_{15} > \frac{2}{10^8}.$$

Also

$$\frac{225 \cdot 15\mu_2^2}{4(1-\gamma_{15})} < \frac{3375}{4} \cdot \frac{16}{10^{12}} \cdot \frac{300000}{299999} < \frac{14}{10^9},$$

where the final cleared-denominator gap is

$$149\,986\,000\,000\,000\,000 > 0.$$

Hence

$$M_{15}^{(1)}(\mu_2, 15) > \frac{2}{10^8} - \frac{14}{10^9} = \frac{6}{10^9} > 0. \quad (6.23)$$

Both cases contradict Lemma 5.6. Thus dimension 15 has no boundary global maximizer.

Theorem 6.4 (Boundary exclusion for $11 \leq n \leq 15$). *For every $n \in \{11, 12, 13, 14, 15\}$, no global maximizer of Φ on K_n has a zero entry.*

Proof. If $\tau = 0$, the contradiction following Lemma 5.5 applies. If $\tau > 0$, Corollary 6.2 handles dimensions 12, 13, 14 and all ordinary pairs in dimension 11; equations (6.11)–(6.14) handle the exceptional pair. The thin–thick argument handles dimension 15. $\square$

7 Boundary exclusion in dimensions eight, nine, and ten

For $n \in \{8, 9, 10\}$, define

$$\begin{array}{c|ccc} n & 8 & 9 & 10 \\ \hline T_n & 1/7 & 1/10 & 1/10 \\ \lambda_n & 20 & 27 & 33. \end{array} \quad (7.1)$$

As before, let A be a hypothetical boundary global maximizer. The constants $\alpha_{n,k}$ satisfy $\alpha_{n,k} \leq n/2$, so Lemma 5.5 gives

$$\tau^2 \leq \frac{n\gamma_n}{1 - \gamma_n} < T_n^2. \quad (7.2)$$

The exact positive gaps in (7.2) are

$$\begin{array}{c|c} n & T_n^2 - \frac{n\gamma_n}{1 - \gamma_n} \\ \hline 8 & \frac{6407093}{7277} \\ 9 & \frac{477848900}{994933} \\ 10 & \frac{156193300}{156193300}. \end{array} \quad (7.3)$$

Lemma 7.1 (Quadratic binomial bounds). *For $n \in \{8, 9, 10\}$ and $0 \leq t \leq T_n$,*

$$(1 - t)^n \geq 1 - nt + \lambda_n t^2. \quad (7.4)$$

Proof. From the binomial expansion,

$$(1 - t)^n \geq 1 - nt + \binom{n}{2} t^2 - \binom{n}{3} t^3.$$

Indeed, after the cubic term the alternating terms have decreasing absolute values on the stated intervals: for $k \geq 3$,

$$\frac{\binom{n}{k+1} t^{k+1}}{\binom{n}{k} t^k} = \frac{n - k}{k + 1} t < 1.$$

Thus the remaining tail starts positive and is nonnegative. Finally,

$$\binom{8}{2} - \binom{8}{3} \frac{1}{7} = 20,$$

$$\binom{9}{2} - \binom{9}{3} \frac{1}{10} = \frac{138}{5} > 27,$$

and

$$\binom{10}{2} - \binom{10}{3} \frac{1}{10} = 33.$$

□

7.1 Two-sided critical pairs

Suppose $a, b \geq 1$. By transposition, assume $a \leq b$. Put

$$p = n - a - b, \quad c_{n,a,b} = \frac{\alpha_{n,a} + \alpha_{n,b}}{p^2},$$

and use the sharp block bound

$$L_{n,a,b}^\sharp = \frac{v_{n-a} v_{n-b}}{v_p}.$$

Lemma 5.7 applies with $\lambda = \lambda_n$.

Proposition 7.2 (Exact two-sided margins). *For every*

$$n \in \{8, 9, 10\}, \quad 1 \leq a \leq b, \quad a + b \leq n - 1,$$

one has

$$M_n^{(2)}(L_{n,a,b}^\sharp, c_{n,a,b}) > 0.$$

The minimum occurs at $(a, b) = (1, 1)$ and equals

n	minimum (decimal for scale)
8	1.379790×10^{-5}
9	5.710428×10^{-6}
10	1.982111×10^{-6} .

(7.5)

Proof. There are respectively 12, 16, 20 unordered pairs. Substitute the exact rational definitions into (5.29), clear denominators, and compare the resulting positive fractions. Appendix B performs exactly this finite calculation. □

Thus no two-sided critical pair can occur.

7.2 One-sided cuts and stationarity

It remains to consider a one-sided critical pair. After transposition, assume $a = 0$. The case $b = 0$ would give $I = J = [n]$ and Hall deficit zero, so a critical pair with $\tau > 0$ necessarily has $1 \leq b \leq n - 1$. Thus $I = [n]$ and J is a set of

$$p = n - b$$

columns satisfying

$$\sum_{j \in J} c_j = p(1 - \tau). \tag{7.6}$$

The corresponding columns of $S = A/(1 - \tau)$ have total sum p . Since $B \leq S$ is doubly stochastic, those same p columns of B also have total sum p . Hence entrywise saturation holds:

$$B[:, J] = S[:, J]. \tag{7.7}$$

Define

$$m_i = \sum_{j \in J} b_{ij}, \quad \beta_i = \frac{m_i}{p}. \quad (7.8)$$

Then $\beta = (\beta_i)$ is a probability vector.

Lemma 7.3 (One-sided stationarity identity). *Put*

$$K = \frac{\Phi(A)}{\prod_i r_i}$$

and

$$D = 1 - \sum_i \frac{\beta_i}{r_i}.$$

Then

$$D = \frac{(K-1)\tau}{1-\tau} \geq \frac{(1-\gamma_n)\tau}{1-\tau} > 0. \quad (7.9)$$

Proof. Apply (5.24) to the p columns in J . Their total mass is $m_E = p(1-\tau)$; the row masses are $\sum_{j \in J} a_{ij}$; each selected column has full column mass; and every permanent monomial uses exactly p selected entries. With

$$u = \sum_i \frac{\sum_{j \in J} a_{ij}}{r_i},$$

the general identity becomes

$$\left(\prod_i r_i \right) u + p \left(\prod_j c_j - \text{per}(A) \right) - p(1-\tau)\Phi(A) = 0.$$

Using $\prod_j c_j - \text{per}(A) = \Phi(A) - \prod_i r_i$,

$$u = p - pK\tau. \quad (7.10)$$

By saturation,

$$\sum_{j \in J} a_{ij} = (1-\tau)m_i.$$

Thus (7.10), divided by p , becomes

$$(1-\tau) \sum_i \frac{\beta_i}{r_i} = 1 - K\tau,$$

which is the asserted identity. Finally,

$$K \geq \Phi(A) \geq 2 - \gamma_n,$$

because $\prod_i r_i \leq 1$. □

7.3 Row-product energy

Define

$$d_0^2 = \frac{n\gamma_n}{2(1-\gamma_n)}. \quad (7.11)$$

For $n = 8, 9, 10$, exact arithmetic gives $d_0 < 1/10$; the positive gaps are

$$\begin{array}{c|c} n & \frac{1}{100} - d_0^2 \\ \hline 8 & \frac{4757}{13075700} \\ 9 & \frac{2762489}{477848900} \\ 10 & \frac{1278433}{156193300} \end{array} \quad (7.12)$$

Lemma 7.4 (Row-product energy). *For $n \in \{8, 9, 10\}$, put*

$$y_i = 1 - \frac{1}{r_i}, \quad C_n = \frac{(1 - \gamma_n)(1 - d_0)^2}{2}.$$

Then

$$\rho \geq C_n \sum_i y_i^2, \quad C_n > \frac{2}{5}. \quad (7.13)$$

In particular, $\rho > \frac{2}{5} \sum_i y_i^2$ whenever $(y_1, \dots, y_n) \neq 0$.

Proof. First, every row sum satisfies $r_i \geq 1 - d_0$. Indeed, if $r_i = 1 - d < 1$, the complementary $n - 1$ rows have positive excess d . Lemma 5.3 and $\alpha_{n,n-1} \leq n/2$ give

$$d^2 \leq \frac{n\rho}{2(1 - \delta)} \leq \frac{n\gamma_n}{2(1 - \gamma_n)} = d_0^2.$$

For every $r \geq 1 - d_0$,

$$r - 1 - \log r \geq \frac{(1 - d_0)^2}{2} \left(1 - \frac{1}{r}\right)^2. \quad (7.14)$$

To see this, set $y = 1 - 1/r$. Then

$$r - 1 - \log r = \int_0^y \frac{t}{(1 - t)^2} dt.$$

For $y \geq 0$, the integrand is at least t ; for $y < 0$, reverse the integral and use $(1 - t)^{-2} \geq (1 - d_0)^2$ on $[y, 0]$.

Since $\sum_i (r_i - 1) = 0$,

$$-\log(1 - \rho) = \sum_i (r_i - 1 - \log r_i) \geq \frac{(1 - d_0)^2}{2} \sum_i y_i^2.$$

Also

$$\rho = 1 - (1 - \rho) \geq (1 - \rho)(-\log(1 - \rho)) \geq (1 - \gamma_n)(-\log(1 - \rho)).$$

For these dimensions, $\gamma_n < 1/81$ and $d_0 < 1/10$, so

$$C_n = \frac{(1 - \gamma_n)(1 - d_0)^2}{2} > \frac{80}{81} \cdot \frac{81}{100} \cdot \frac{1}{2} = \frac{2}{5}.$$

This proves the non-strict C_n bound. If $y \neq 0$, then $\sum_i y_i^2 > 0$ and the strict comparison with $2/5$ follows. $\square$

Let

$$s = \sum_i \left(\beta_i - \frac{1}{n} \right)^2. \quad (7.15)$$

By the harmonic-mean inequality,

$$\sum_i \frac{1}{r_i} \geq \frac{n^2}{\sum_i r_i} = n,$$

so $\sum_i y_i \leq 0$. Therefore

$$D = \sum_i \beta_i y_i \leq \sum_i \left(\beta_i - \frac{1}{n} \right) y_i \leq \sqrt{s} \left(\sum_i y_i^2 \right)^{1/2}.$$

Since $D > 0$, one has $s > 0$ and $y \neq 0$. Thus the strict form of Lemma 7.4 applies, and (7.9) implies

$$\rho > \frac{2}{5} \frac{D^2}{s} \geq \frac{2}{5} \frac{(1 - \gamma_n)^2 \tau^2}{(1 - \tau)^2 s}. \quad (7.16)$$

The p columns in J are probability vectors whose average is β . Convexity of squared Euclidean distance shows that at least one of them, say $b^{(j)}$, satisfies

$$s(b^{(j)}) \geq s.$$

Theorem 3.9 and Corollary 3.6 give

$$\text{per}(B) \geq \max\{\kappa_n, \gamma_n e^{s/2}\}. \quad (7.17)$$

Consequently,

$$\text{per}(A) \geq (1 - \tau)^n \max\{\kappa_n, \gamma_n e^{s/2}\}. \quad (7.18)$$

On the other hand, $\delta \geq \rho$, and therefore

$$\text{per}(A) = \gamma_n - \delta \leq \gamma_n - \rho. \quad (7.19)$$

For a one-sided pair, (5.20), $p \geq 1$, and $\alpha_{n,b} \leq n/2$ give

$$\tau^2 \leq \frac{n\gamma_n}{2(1 - \gamma_n)} = d_0^2 < \frac{1}{100}. \quad (7.20)$$

7.4 The two entropy regimes

Set

$$s_0 = 2 \log \frac{\kappa_n}{\gamma_n} > 0.$$

Lemma 7.5 (Small entropy). *If $s \leq s_0$, then*

$$\rho + \kappa_n(1 - \tau)^n > \gamma_n. \quad (7.21)$$

Proof. From (7.16),

$$\rho > Q_n \tau^2, \quad Q_n = \frac{2(1 - \gamma_n)^2}{5s_0}.$$

Bernoulli's inequality gives

$$\rho + \kappa_n(1 - \tau)^n - \gamma_n > \kappa_n - \gamma_n - n\kappa_n\tau + Q_n\tau^2.$$

The right side is at least

$$\kappa_n - \gamma_n - \frac{n^2 \kappa_n^2}{4Q_n}.$$

Since $\log x \leq x - 1$,

$$s_0 \leq \frac{2(\kappa_n - \gamma_n)}{\gamma_n}.$$

Thus the last expression is positive whenever

$$\frac{2}{5}(1 - \gamma_n)^2 \gamma_n - \frac{n^2}{2} \kappa_n^2 > 0. \quad (7.22)$$

The exact positive values of the left side are

n	value (decimal for scale)
8	7.682036×10^{-4}
9	3.378893×10^{-4}
10	1.383834×10^{-4} .

(7.23)

□

Lemma 7.6 (Large entropy). *If $s \geq s_0$, then*

$$\rho + \gamma_n(1 - \tau)^n e^{s/2} > \gamma_n. \quad (7.24)$$

Proof. Put $q = (1 - \tau)^n$. Since $e^{s/2} \geq 1 + s/2$, equation (7.16) and AM–GM give

$$\begin{aligned} \rho + \gamma_n q e^{s/2} - \gamma_n &> \frac{2}{5} \frac{D^2}{s} + \gamma_n q \left(1 + \frac{s}{2}\right) - \gamma_n \\ &\geq D \sqrt{\frac{4}{5} \gamma_n q} + \gamma_n (q - 1). \end{aligned}$$

By (7.9), (7.20), and Bernoulli's inequality,

$$D \sqrt{q} \geq (1 - \gamma_n) \tau \left(\frac{9}{10}\right)^{n/2}, \quad q - 1 \geq -n\tau.$$

Therefore the preceding expression is at least

$$\tau \left[(1 - \gamma_n) \left(\frac{9}{10}\right)^{n/2} \sqrt{\frac{4}{5} \gamma_n} - n\gamma_n \right].$$

The bracket is positive exactly when

$$\frac{4}{5}(1 - \gamma_n)^2 \left(\frac{9}{10}\right)^n - n^2 \gamma_n > 0. \quad (7.25)$$

The exact positive values are

n	value (decimal for scale)
8	1.889119×10^{-1}
9	2.334869×10^{-1}
10	2.424523×10^{-1} .

(7.26)

□

Equations (7.18) and (7.19) contradict Lemma 7.5 when $s \leq s_0$ and Lemma 7.6 when $s \geq s_0$. Thus one-sided critical pairs are impossible.

Theorem 7.7 (Boundary exclusion for $8 \leq n \leq 10$). *For every $n \in \{8, 9, 10\}$, no global maximizer of Φ on K_n has a zero entry.*

Proof. The case $\tau = 0$ was excluded after Lemma 5.5. For $\tau > 0$, Proposition 7.2 excludes two-sided critical pairs, and the one-sided analysis above excludes the remaining pairs. □

8 Boundary exclusion in dimension seven

Dimension seven is the first dimension in which the scalar zero-block comparison alone fails. The failure is confined to critical Hall pairs with $p = n - a - b = 1$. The complementary-minor entropy from Theorem 3.12 supplies exactly the missing information.

Throughout this section let $n = 7$, write

$$\gamma = \gamma_7 = \frac{720}{117649},$$

and let A be a hypothetical boundary global maximizer. The localization parameter is

$$\eta_7 = \frac{1}{47}.$$

The two inequalities needed in Lemma 5.3 are

$$\eta_7^2 - \frac{\gamma}{14(1-\gamma)} = \frac{23263}{1808073127} > 0, \quad \frac{1}{2} - \frac{3}{7} - \eta_7 = \frac{33}{658} > 0. \quad (8.1)$$

Thus all subset-excess estimates of Section 5 apply in dimension seven. In particular,

$$\max_{1 \leq k \leq 6} \alpha_{7,k} = \alpha_{7,3} = \frac{53576}{15463} < \frac{7}{2}. \quad (8.2)$$

As before, the case $\tau = 0$ is impossible, so fix a critical pair and use a, b, p, e, f, x as in Lemma 5.5.

8.1 Two-sided pairs with core size at least two

Put

$$T_7 = \frac{5}{24}, \quad \lambda_7 = \frac{329}{24} = \binom{7}{2} - \binom{7}{3} T_7. \quad (8.3)$$

The common Hall estimate and (8.2) imply

$$\tau^2 \leq \frac{7\gamma}{1-\gamma} < T_7^2, \quad T_7^2 - \frac{7\gamma}{1-\gamma} = \frac{20185}{67351104} > 0. \quad (8.4)$$

The alternating binomial argument of Lemma 7.1 gives

$$(1-t)^7 \geq 1 - 7t + \lambda_7 t^2 \quad (0 \leq t \leq T_7). \quad (8.5)$$

Proposition 8.1 (Ordinary dimension-seven margins). *For every $1 \leq a \leq b$ with $p = 7 - a - b \geq 2$,*

$$M_7^{(2)} \left(L_{7,a,b}^\sharp, \frac{\alpha_{7,a} + \alpha_{7,b}}{p^2} \right) > 0,$$

where (5.29) is evaluated with $\lambda = 329/24$. There are six pairs, and the minimum is attained at $(a, b) = (1, 1)$:

$$\frac{217499955444827479495}{21713873952987016007066496} > 0 \quad (\approx 1.001664 \times 10^{-5}). \quad (8.6)$$

Proof. This is an exact rational substitution in Lemma 5.7. The six comparisons are checked by the supplementary certificate described in Appendix B. $\square$

Proposition 8.1 excludes every two-sided critical pair with $p \geq 2$.

8.2 One-sided pairs

The one-sided stationarity identity from Lemma 7.3 remains valid. We sharpen only the numerical energy estimates. Define

$$d_7^2 = \frac{7\gamma}{2(1-\gamma)} = \frac{2520}{116929}.$$

Then

$$\frac{9}{400} - d_7^2 = \frac{44361}{46771600} > 0, \quad (8.7)$$

so every row sum and every column sum of A is greater than $17/20$.

Lemma 8.2 (Dimension-seven product energy). *With $y_i = 1 - r_i^{-1}$, put*

$$C_7 = \frac{(1-\gamma)(17/20)^2}{2}.$$

Then

$$\rho \geq C_7 \sum_i y_i^2, \quad C_7 > \frac{1}{3}. \quad (8.8)$$

Thus $\rho > \frac{1}{3} \sum_i y_i^2$ unless all row sums equal one. The transposed statement holds for the column sums.

Proof. The proof of Lemma 7.4 gives the stated non-strict bound with coefficient C_7 . This coefficient exceeds $1/3$, since

$$\frac{(1-\gamma)(17/20)^2}{2} - \frac{1}{3} = \frac{7258243}{282357600} > 0.$$

□

Retain the notation $D, s, q = (1-\tau)^7$ from the one-sided analysis in Section 7. Here $D > 0$, so the reciprocal-deviation vector y is nonzero and the strict $1/3$ form of Lemma 8.2 applies. Hence

$$\rho > \frac{D^2}{3s} \geq \frac{(1-\gamma)^2 \tau^2}{3(1-\tau)^2 s}. \quad (8.9)$$

Also $\tau < 3/20$ by (8.7).

Proposition 8.3 (One-sided exclusion in dimension seven). *No one-sided critical pair occurs at a boundary global maximizer in dimension seven.*

Proof. Set $s_0 = 2 \log(\kappa_7/\gamma)$. If $s \leq s_0$, the proof of Lemma 7.5, with $1/3$ in place of $2/5$, reduces the contradiction to

$$\frac{1}{3}(1-\gamma)^2 \gamma - \frac{49}{2} \kappa_7^2 = \frac{22176489486262543850191055}{20672701805255574480144334848} > 0. \quad (8.10)$$

Suppose $s \geq s_0$. From (8.9), $e^{s/2} \geq 1 + s/2$, and AM–GM,

$$\rho + \gamma q e^{s/2} - \gamma > D \sqrt{\frac{2\gamma q}{3}} + \gamma(q-1). \quad (8.11)$$

The stationarity bound gives

$$D\sqrt{q} \geq (1-\gamma)\tau(1-\tau)^{5/2}.$$

Moreover, for $0 \leq t \leq 3/20$,

$$\frac{1 - (1 - t)^7}{t} \leq 7 - 21t + 35t^2.$$

The preceding estimates naturally reduce positivity to

$$\frac{2}{3}(1 - \gamma)^2(1 - t)^5 > \gamma \left(\frac{1 - (1 - t)^7}{t} \right)^2.$$

Since $0 \leq t \leq 3/20$, one has $(1 - t)^5 \geq (1 - t)^6$. It is therefore sufficient to verify the stronger inequality obtained by replacing the left factor $(1 - t)^5$ by the smaller $(1 - t)^6$. Together with the displayed quadratic upper bound, this gives the sufficient condition

$$W(t) := \frac{2}{3}(1 - \gamma)^2(1 - t)^6 - \gamma(7 - 21t + 35t^2)^2 > 0. \quad (8.12)$$

The degree-six Bernstein coefficients of W on $[0, 3/20]$ are all positive; the smallest is

$$\frac{155130361249919329}{1328763571296000000} > 0. \quad (8.13)$$

This exact coefficient calculation appears in the supplementary certificate. Thus both entropy regimes contradict the upper bound $\text{per}(A) \leq \gamma - \rho$, exactly as in Section 7. $\square$

8.3 Single-bridge pairs: preliminary reductions

It remains to treat $p = 1$. By transposition assume $a \leq b$, so

$$(a, b) \in \{(1, 5), (2, 4), (3, 3)\}. \quad (8.14)$$

Recall

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c),$$

so criticality gives

$$\tau = e + f - x. \quad (8.15)$$

If $e \leq 0$, then $\tau \leq f_+$ and the proof of Lemma 5.5 gives the sharper estimate

$$\tau^2 \leq \alpha_{7,b} \frac{\delta}{1 - \delta}.$$

If $f \leq 0$, the analogous estimate uses $\alpha_{7,a}$. Since

$$\left(\frac{3}{20} \right)^2 - \alpha_{7,3} \frac{\gamma}{1 - \gamma} = \frac{842770143}{723229250800} > 0, \quad (8.16)$$

we may use the quadratic binomial bound with

$$T = \frac{3}{20}, \quad \lambda = \binom{7}{2} - \binom{7}{3} T = \frac{63}{4}.$$

Proposition 8.4 (Sign-mixed single-bridge margins). *For the three pairs in (8.14), all six applications of Lemma 5.7 obtained from $e \leq 0$ or $f \leq 0$ have positive margin. Their minimum is*

$$\frac{165035926385}{3897251304526692} > 0 \quad (\approx 4.234163 \times 10^{-5}), \quad (8.17)$$

attained for $(a, b) = (1, 5)$ with $e \leq 0$.

Proof. Substitute $L = L_{7,a,b}^\sharp$ and respectively $c = \alpha_{7,b}$ or $c = \alpha_{7,a}$ into (5.29), with $\lambda = 63/4$. The supplementary script checks the six exact fractions. $\square$

We may henceforth assume

$$e > 0, \quad f > 0. \quad (8.18)$$

The subset estimates and (8.2) imply

$$0 < e, f < \frac{3}{20}. \quad (8.19)$$

Put

$$t = e + f, \quad m = 7 - a = b + 1, \quad \ell = 7 - b = a + 1, \quad d = \min\{a, b\} = a.$$

Define the two group-product envelopes

$$P_a(e) = \left(1 + \frac{e}{a}\right)^a \left(1 - \frac{e}{7-a}\right)^{7-a}, \quad P_b(f) = \left(1 + \frac{f}{b}\right)^b \left(1 - \frac{f}{7-b}\right)^{7-b}. \quad (8.20)$$

If

$$P = \prod_i r_i, \quad Q = \prod_j c_j, \quad H = \text{per}(A),$$

write

$$u_r = P_a(e) - P, \quad u_c = P_b(f) - Q,$$

and set

$$D_0 = 2 - P_a(e) - P_b(f), \quad K = \gamma - D_0, \quad u = \delta - D_0. \quad (8.21)$$

Then

$$u \geq u_r + u_c \geq 0, \quad H = K - u. \quad (8.22)$$

Lemma 8.5 (Conditional product energy). *Let*

$$V_I = \sum_{i \in I} \left(\frac{1}{r_i} - \frac{1}{m} \sum_{h \in I} \frac{1}{r_h} \right)^2$$

and define V_J analogously from the columns indexed by J . Then

$$V_I \leq 3u_r, \quad V_J \leq 3u_c. \quad (8.23)$$

Proof. All row and column sums exceed $r_0 = 17/20$. Let a group of k positive numbers z_i have arithmetic mean μ . Since $\sum_i (z_i/\mu - 1) = 0$,

$$k \log \mu - \sum_i \log z_i = \sum_i \left(\frac{z_i}{\mu} - 1 - \log \frac{z_i}{\mu} \right).$$

For completeness, set $y = 1 - \mu/z_i$. Then

$$\frac{z_i}{\mu} - 1 - \log \frac{z_i}{\mu} = \int_0^y \frac{t}{(1-t)^2} dt, \quad \frac{1}{z_i} - \frac{1}{\mu} = -\frac{y}{\mu}.$$

If $y \geq 0$, the integral is at least $y^2/2$, while $r_0^2/\mu^2 \leq 1$. If $y < 0$, then on $[y, 0]$,

$$(1-t)^{-2} \geq (1-y)^{-2} = \left(\frac{z_i}{\mu} \right)^2 \geq \frac{r_0^2}{\mu^2}.$$

Both cases give

$$\frac{z_i}{\mu} - 1 - \log \frac{z_i}{\mu} \geq \frac{r_0^2}{2} \left(\frac{1}{z_i} - \frac{1}{\mu} \right)^2. \quad (8.24)$$

Applying this to the two row groups and using the fact that variance is minimized at the arithmetic mean of the reciprocals,

$$\log \frac{P_a(e)}{P} \geq \frac{r_0^2}{2} V_I.$$

Therefore

$$u_r = P_a(e) - P = P \left(\exp \left(\log \frac{P_a(e)}{P} \right) - 1 \right) \geq (1 - \gamma) \frac{r_0^2}{2} V_I \geq \frac{1}{3} V_I,$$

where the last inequality is the exact comparison in Lemma 8.2. The column proof is identical. $\square$

8.4 Core stationarity and the entropy squeeze

Put $\theta = 1 - \tau$. At a critical pair, the doubly stochastic matrix $B \leq A/\theta$ has $B[I^c, J^c] = 0$. The doubly stochastic cut identity shows that $B[I, J]$ has total mass one. Criticality and (8.15) show that $(A/\theta)[I, J]$ has the same total mass. Since the first matrix is entrywise dominated by the second, equality of their total masses forces termwise saturation:

$$B[I, J] = \frac{A[I, J]}{\theta}. \quad (8.25)$$

Let ξ and ζ be the row- and column-sum vectors of this core and put

$$s_I = \sum_{i \in I} \left(\xi_i - \frac{1}{m} \right)^2, \quad s_J = \sum_{j \in J} \left(\zeta_j - \frac{1}{\ell} \right)^2, \quad s = s_I + s_J. \quad (8.26)$$

For $k \geq 0$, let H_k be the contribution to H of permutations using exactly k entries of the block $I^c \times J^c$. Put H_0 for the zero-use contribution.

Lemma 8.6 (Single-bridge stationarity). *One has*

$$P \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + Q \left(\sum_{j \in J} \frac{\zeta_j}{c_j} - 1 \right) = \frac{\tau H + \sum_{k \geq 1} k H_k}{1 - \tau}. \quad (8.27)$$

Moreover,

$$H_0 \geq L_{\tau, a, b}^\sharp (1 - t)^7 e^{s/2}. \quad (8.28)$$

Proof. Apply the common identity (5.24) to the core $E = I \times J$. Its total mass is θ ; by (8.25), its row and column masses are $\theta \xi_i$ and $\theta \zeta_j$. A permutation counted by H_k uses exactly $k + 1$ core entries. Substitution in (5.24), followed by division by θ and use of $\Phi = P + Q - H$, gives exactly (8.27).

For the second assertion, every permanent term of B avoids the prescribed zero block. Entrywise domination $A \geq \theta B$ therefore maps each term of $\text{per}(B)$ to an H_0 term with at least θ^7 times its weight. Thus, term by term,

$$H_0 \geq \theta^7 \text{per}(B) \geq L_{\tau, a, b}^\sharp \theta^7 e^{s/2},$$

where the last inequality is Theorem 3.12. Finally $\theta = 1 - e - f + x \geq 1 - t$. $\square$

For brevity write

$$L = L_{7,a,b}^\sharp, \quad A_0 = L(1-t)^7, \quad (8.29)$$

and define

$$\mathcal{B} = P_a(e) \frac{e}{m-e} + P_b(f) \frac{f}{\ell-f} - \frac{(t+d)K - dA_0}{1-t}. \quad (8.30)$$

Lemma 8.7 (Stationarity–entropy squeeze). *At a feasible single-bridge maximizer satisfying (8.18), one has*

$$H \geq A_0 e^{s/2}, \quad (8.31)$$

and, whenever $\mathcal{B} > 0$,

$$su \geq \frac{\mathcal{B}^2}{3}. \quad (8.32)$$

On the other hand,

$$su \leq \frac{(K - A_0)^2}{2A_0}. \quad (8.33)$$

Proof. The first assertion is (8.28) and $H \geq H_0$. By the harmonic-mean inequality,

$$\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \geq \frac{e}{m-e} - \sqrt{s_I V_I},$$

and similarly in the column group. Since a permutation uses at most d entries of $I^c \times J^c$,

$$\sum_{k \geq 1} k H_k \leq d(H - H_0).$$

The right side of (8.27) is therefore at most

$$\frac{(\tau+d)(K-u) - dA_0}{1-\tau} \leq \frac{(t+d)(K-u) - dA_0}{1-t}. \quad (8.34)$$

Indeed, the derivative of the first rational function with respect to τ is

$$\frac{(1+d)(K-u) - dA_0}{(1-\tau)^2} > 0,$$

because $K - u = H \geq H_0 \geq A_0$, and then $\tau \leq t$.

Put

$$E = \frac{e}{m-e}, \quad F = \frac{f}{\ell-f}, \quad c = \frac{t+d}{1-t},$$

and denote the common value in (8.27) by Z . The harmonic-mean estimates and $P, Q \leq 1$ give

$$PE + QF - Z \leq \sqrt{s_I V_I} + \sqrt{s_J V_J}. \quad (8.35)$$

On the other hand, (8.34) and $P = P_a - u_r$, $Q = P_b - u_c$ give

$$PE + QF - Z \geq \mathcal{B} + cu - u_r E - u_c F.$$

Here $c \geq \max\{E, F\}$, because $E \leq e/(1-e) \leq t/(1-t) < c$, and similarly for F . Together with $u \geq u_r + u_c$, this proves $PE + QF - Z \geq \mathcal{B}$. Lemma 8.5 and Cauchy–Schwarz now yield

$$\mathcal{B} \leq \sqrt{s_I V_I} + \sqrt{s_J V_J} \leq \sqrt{3s(u_r + u_c)} \leq \sqrt{3su}.$$

This proves (8.32).

Finally, $K - u = H \geq A_0 e^{s/2}$ and $e^{s/2} \geq 1 + s/2$ imply

$$su \leq \frac{2u(K - A_0 - u)}{A_0} \leq \frac{(K - A_0)^2}{2A_0},$$

which is (8.33). □

8.5 The exact two-variable certificate

Lemma 8.8 (Exact Bernstein certificate principle). *Let P be a rational polynomial on a rational box. Express P in the tensor-product Bernstein basis of its coordinatewise degree. If every Bernstein coefficient is positive, then P is positive on the box. Moreover, exact de Casteljau subdivision at a rational parameter replaces a parent box by finitely many child boxes whose union is the parent and whose Bernstein coefficients are exact rational affine combinations of the parent coefficients. Consequently, a finite subdivision tree in which every leaf has positive Bernstein coefficients is an exact certificate that $P > 0$ on the original box.*

Proof. Each tensor-product Bernstein basis function is nonnegative on the box and the basis functions sum to one, so the value of P is a convex combination of its Bernstein coefficients. The de Casteljau identities are polynomial identities and partition each coordinate interval exactly; iterating them therefore covers the original box without gaps or overlaps except along shared boundaries. Rational subdivision points preserve exact rational arithmetic. $\square$

All quantities in the preceding subsection are rational functions of e, f . The actual feasible set is contained in the larger closed square

$$\mathcal{Q} = \left[0, \frac{3}{20}\right]^2.$$

We certify the entire square, so no separate semialgebraic description of the transportation-feasible subset is required. On $\mathcal{Q}$ one has

$$m - e > 0, \quad \ell - f > 0, \quad 1 - e - f \geq \frac{7}{10} > 0,$$

and also $A_0 = L(1 - e - f)^7 > 0$. Hence

$$\Delta = (m - e)(\ell - f)(1 - e - f) > 0,$$

so clearing denominators preserves every inequality direction. Define the three rational-coefficient polynomials

$$\mathcal{G} = A_0 - K, \quad \mathcal{N} = \Delta \mathcal{B},$$

and

$$\mathcal{F} = 2A_0\mathcal{N}^2 - 3(K - A_0)^2\Delta^2. \tag{8.36}$$

If $\mathcal{G} > 0$, then (8.31) contradicts $H = K - u \leq K$. If $\mathcal{N} > 0$ and $\mathcal{F} > 0$, then (8.32) and (8.33) contradict each other. Thus it remains only to certify the following finite polynomial alternative.

Proposition 8.9 (Exact single-bridge certificate). *For each $(a, b) \in \{(1, 5), (2, 4), (3, 3)\}$ and every $(e, f) \in \mathcal{Q}$,*

$$\mathcal{G}(e, f) > 0 \quad \text{or} \quad (\mathcal{N}(e, f) > 0 \text{ and } \mathcal{F}(e, f) > 0). \tag{8.37}$$

Proof. Apply Lemma 8.8. Starting from $\mathcal{Q}$, the supplementary script subdivides the physically longer side at its midpoint whenever neither branch in (8.37) is yet coefficientwise positive. All coefficients and all subdivision endpoints are instances of `fractions.Fraction`; no floating-point operation or tolerance is used.

The resulting finite trees are

(a, b)	nodes	$\mathcal{G}$ -leaves	$(\mathcal{N}, \mathcal{F})$ -leaves	maximum depth
(1, 5)	39	10	10	8
(2, 4)	35	9	9	6
(3, 3)	47	11	13	8.

(8.38)

The complete coefficient generation and de Casteljau subdivision are given in the supplementary certificate described in Appendix B. $\square$

Theorem 8.10 (Boundary exclusion in dimension seven). *No global maximizer of Φ on K_7 has a zero entry.*

Proof. The case $\tau = 0$ is excluded after Lemma 5.5. For $\tau > 0$, Proposition 8.3 excludes one-sided pairs, Proposition 8.1 excludes two-sided pairs with $p \geq 2$, and Proposition 8.4 excludes single-bridge pairs with a nonpositive complementary excess. In the remaining case, Lemma 8.7 and Proposition 8.9 give a contradiction. Hence no boundary global maximizer exists. $\square$

9 Boundary exclusion in dimension six

We now treat the remaining dimension. Throughout this section,

$$n = 6, \quad \gamma = \gamma_6 = \frac{5}{324}, \quad \kappa = \kappa_6 = v_5 G(5) = \frac{6144}{390625} > \gamma. \quad (9.1)$$

Let A be a hypothetical boundary global maximizer, and retain the notation

$$R = \prod_i r_i = 1 - \rho, \quad C = \prod_j c_j = 1 - \sigma, \quad H = \text{per}(A) = \gamma - \delta$$

from Section 5.

9.1 Uniform localization

Lemma 9.1 (Dimension-six localization). *For every nonempty proper set E of rows, if*

$$\sum_{i \in E} r_i = |E| + e,$$

then

$$|e|^2 \leq \frac{15}{319} < \left(\frac{7}{32}\right)^2 < \left(\frac{11}{50}\right)^2. \quad (9.2)$$

The same assertion holds for every column subset. In particular,

$$r_i, c_j > \frac{25}{32} \quad (1 \leq i, j \leq 6). \quad (9.3)$$

Proof. If $e > 0$, separate AM–GM on E and E^c , followed by binary Pinsker, gives

$$\frac{2e^2}{6} \leq -\log(1 - \rho) \leq \frac{\rho}{1 - \rho} \leq \frac{\gamma}{1 - \gamma}.$$

Thus $e^2 \leq 15/319$. If $e < 0$, the complementary row set has positive excess $-e$. The column proof is identical. The rational comparisons in (9.2) are verified exactly in the supplementary certificate. $\square$

Set

$$E_0 = \frac{11}{50}. \quad (9.4)$$

As in the preceding dimensions, a boundary maximizer with Hall deficit $\tau = 0$ is already doubly stochastic and has a zero. Corollary 3.6 would then give $H \geq \kappa > \gamma$, contradicting Lemma 5.1. Hence $\tau > 0$.

9.2 Two-sided critical Hall pairs

Assume first that a critical pair has $a, b \geq 1$, and transpose so that $a \leq b$. The six possible types are

$$(a, b, p) = (1, 1, 4), (1, 2, 3), (1, 3, 2), (2, 2, 2), (1, 4, 1), (2, 3, 1), \quad (9.5)$$

where $p = 6 - a - b$. Put

$$e = \sum_{i \in I^c} r_i - a, \quad f = \sum_{j \in J^c} c_j - b, \quad x = s_A(I^c, J^c), \quad t = e + f. \quad (9.6)$$

Criticality gives

$$p\tau = e + f - x = t - x. \quad (9.7)$$

Consequently,

$$-E_0 \leq e, f \leq E_0, \quad 0 < t \leq 2E_0 = \frac{11}{25} < p, \quad \tau \leq \frac{t}{p}. \quad (9.8)$$

Put $\theta = 1 - \tau$. The doubly stochastic cut identity and the forced zero block give $s_B(I, J) = p$. Criticality gives $s_A(I, J) = p\theta$, so $B[I, J] \leq (A/\theta)[I, J]$ and both sides have total mass p . Hence entrywise domination and equality of total masses imply the termwise saturation

$$B[I, J] = \frac{A[I, J]}{\theta}.$$

Let $m = 6 - a$, $\ell = 6 - b$, and let ξ, η be the normalized row and column marginals of this core, as in Proposition 3.14. Put

$$S_\xi = \sum_{i \in I} \left(\xi_i - \frac{1}{m} \right)^2, \quad S_\eta = \sum_{j \in J} \left(\eta_j - \frac{1}{\ell} \right)^2. \quad (9.9)$$

Decompose the permanent as

$$H = H_0 + H_1 + \cdots + H_d, \quad d = \min(a, b) = a,$$

where H_k is the contribution of permutations using exactly k entries of $I^c \times J^c$. Such a permutation uses exactly $p + k$ core entries.

Lemma 9.2 (Dimension-six core stationarity). *At the global maximizer,*

$$R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right) = \frac{\tau H + p^{-1} \sum_{k \geq 1} k H_k}{1 - \tau}. \quad (9.10)$$

Consequently,

$$R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) + C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right) \leq \frac{(t + d)H - dH_0}{p - t}. \quad (9.11)$$

Proof. Apply (5.24) to the core $I \times J$. Its mass is $p\theta$; its row and column masses are $p\theta\xi_i$ and $p\theta\eta_j$; and a monomial in H_k uses exactly $p + k$ core entries. Substitution in the common identity, division by $p\theta$, and $\Phi = R + C - H$ give (9.10) term by term.

Since $k \leq d$,

$$\sum_{k \geq 1} k H_k \leq d(H - H_0).$$

Writing $z = p\tau$, the resulting upper bound is

$$f(z) = \frac{(z + d)H - dH_0}{p - z}.$$

Its derivative has numerator

$$(p + d)H - dH_0 \geq pH > 0.$$

Here $H > 0$ because $A \geq \theta B$, B is doubly stochastic, and hence $H \geq \theta^6 \text{per}(B) \geq \theta^6 \gamma_6 > 0$. Thus f is strictly increasing. Since $z = p\tau = t - x \leq t$, substituting t proves (9.11). $\square$

Let $L = L_{6,a,b}^\sharp$. Define c_ξ, c_η by

$$(c_\xi, c_\eta) = \begin{cases} \left(\frac{p^2}{2(a^2 + b^2)}, \frac{p^2}{2(a^2 + b^2)} \right), & p \geq 2, \\ \left(\frac{1}{2} + \frac{1}{3m}, \frac{1}{2} + \frac{1}{3\ell} \right), & p = 1. \end{cases} \quad (9.12)$$

Put

$$A_0 = L \left(1 - \frac{t}{p} \right)^6. \quad (9.13)$$

Every permanent term of B avoids the forced zero block. Since $A \geq \theta B$, termwise domination gives $H_0 \geq \theta^6 \text{per}(B)$. Proposition 3.14 and $\theta = 1 - \tau \geq 1 - t/p$ therefore give

$$H_0 \geq A_0(1 + c_\xi S_\xi + c_\eta S_\eta). \quad (9.14)$$

The joint deficit budget also gives

$$H \leq \overline{H} := R + C - 2 + \gamma. \quad (9.15)$$

It follows from Lemma 9.2 that both

$$F = \frac{(t + d)\overline{H} - dA_0(1 + c_\xi S_\xi + c_\eta S_\eta)}{p - t} - R \left(\sum_{i \in I} \frac{\xi_i}{r_i} - 1 \right) - C \left(\sum_{j \in J} \frac{\eta_j}{c_j} - 1 \right), \quad (9.16)$$

$$G = \overline{H} - A_0(1 + c_\xi S_\xi + c_\eta S_\eta) \quad (9.17)$$

are nonnegative. Indeed, in (9.11) replace H by the larger quantity $\overline{H}$ and H_0 by the lower bound (9.14); this gives $F \geq 0$. Moreover $\overline{H} \geq H \geq H_0$, which gives $G \geq 0$. Hence $F + \zeta G \geq 0$ for every $\zeta \geq 0$.

9.3 A two-level extremal reduction

For a group of r nonnegative numbers z_i with total s , a probability vector w , and constants $K, B_*, \lambda > 0$, define on the closed product of simplices the genuine polynomial

$$\mathcal{T}(z, w) = KB_* \prod_{i=1}^r z_i - K \sum_{i=1}^r w_i \prod_{j \neq i} z_j - \lambda \sum_{i=1}^r \left(w_i - \frac{1}{r} \right)^2. \quad (9.18)$$

When every $z_i > 0$, this is equal to

$$K \prod_i z_i \left(B_* - \sum_i \frac{w_i}{z_i} \right) - \lambda \sum_i \left(w_i - \frac{1}{r} \right)^2.$$

Lemma 9.3 (Two-level side maximizer). *Suppose $s/r > 25/32$ and $B_* > 32/25$. The maximum of (9.18) is attained in the following form. For some $k \in \{1, \dots, r\}$,*

$$w_i = \frac{1}{k} \quad (1 \leq i \leq k), \quad w_i = 0 \quad (i > k),$$

and the z_i have two levels,

$$z_i = h \quad (i \leq k), \quad z_i = \ell_0 \quad (i > k), \quad h \geq \ell_0.$$

For $k < r$ they can be parametrized by $0 \leq u \leq 1$ as

$$h = \frac{s}{r} \left(1 + u \frac{r-k}{k} \right), \quad \ell_0 = \frac{s}{r} (1 - u). \quad (9.19)$$

For $k = r$, one has $h = s/r$.

Proof. Compactness gives a maximizer. If some $z_i = 0$, the first term in the polynomial form (9.18) vanishes; the second and third terms are nonpositive. By contrast, at $z_i = s/r$ and $w_i = 1/r$ the value is

$$K(s/r)^r (B_* - r/s) > 0,$$

because $r/s < 32/25 < B_*$. Hence every maximizing z_i is positive, the maximum is positive, and at a maximizer $B_* - \sum_i w_i/z_i > 0$.

Among all maximizers choose one minimizing

$$\mathcal{V} = \sum_i z_i^2 + \sum_i w_i^2.$$

Fix $i \neq j$ and preserve the two z - and w -sums by writing

$$z_i = h + x, \quad z_j = h - x, \quad w_i = \omega + y, \quad w_j = \omega - y.$$

Set

$$P_0 = K \prod_{\nu \neq i,j} z_\nu > 0, \quad A = B_* - \sum_{\nu \neq i,j} \frac{w_\nu}{z_\nu} > 0.$$

Expanding the polynomial (9.18), all terms depending on (x, y) are, up to an additive constant,

$$-P_0 A x^2 + 2P_0 x y - 2\lambda y^2. \quad (9.20)$$

This explicit expansion is the reason for using the polynomial definition on the closed domain.

Assume first that $w_i, w_j > 0$. Then (x, y) lies in the relative interior of the feasible pair rectangle, so both stationarity equations apply:

$$-2P_0 A x + 2P_0 y = 0, \quad 2P_0 x - 4\lambda y = 0.$$

Either $x = y = 0$, or $y = Ax$ and $P_0 = 2\lambda A$. In the latter case (9.20) is

$$-\frac{P_0}{A} (Ax - y)^2.$$

For $0 \leq \theta \leq 1$, replacing (x, y) by $(\theta x, \theta y)$ keeps $z_i, z_j > 0$ and $w_i, w_j \geq 0$, preserves both pair sums, and remains on this flat line. It therefore preserves the maximum, whereas

$$\mathcal{V}(\theta) - \mathcal{V}(0) = 2\theta^2 (x^2 + y^2)$$

is strictly positive for $\theta > 0$ unless $x = y = 0$. This contradicts the minimal choice of $\mathcal{V}$. Thus all positive coordinates of w are equal, and their corresponding z_i are equal.

If $w_i = w_j = 0$, then $y = 0$ and the pair-dependent polynomial is the strictly concave quadratic $-P_0 A x^2$. Hence all inactive z_i are equal. Since w is a probability vector, if its support has size k , every active coordinate equals $1/k$.

It remains to order the two z -levels. Suppose $k < r$, transfer a small amount of mass from one active w -coordinate to one inactive coordinate, and keep z fixed. This is a feasible right-sided perturbation. Its directional derivative at zero is

$$K \prod_i z_i \left(\frac{1}{h} - \frac{1}{\ell_0} \right) + \frac{2\lambda}{k} \leq 0.$$

If $h \leq \ell_0$, the left side is strictly positive, a contradiction. Thus $h > \ell_0$ when $k < r$; for $k = r$ there is only one level. Solving the fixed-total equation $kh + (r - k)\ell_0 = s$ yields (9.19), with $0 \leq u \leq 1$. $\square$

Set

$$\zeta = \begin{cases} 14, & p \geq 2, \\ 8, & p = 1, \end{cases} \quad (9.21)$$

and define

$$B_* = \frac{p+d}{p-t} + \zeta, \quad \lambda_\xi = c_\xi A_0 \left(\frac{d}{p-t} + \zeta \right), \quad \lambda_\eta = c_\eta A_0 \left(\frac{d}{p-t} + \zeta \right). \quad (9.22)$$

By (9.3), $\sum_i \xi_i / r_i < 50/39 < B_*$, and similarly for columns. AM–GM on the complementary groups therefore increases the upper bound for $F + \zeta G$ when R, C are replaced by

$$\hat{R} = \left(1 + \frac{e}{a} \right)^a \prod_{i \in I} r_i, \quad \hat{C} = \left(1 + \frac{f}{b} \right)^b \prod_{j \in J} c_j.$$

Consequently,

$$F + \zeta G \leq \mathcal{U} = C_0(t) + T_a(e, r, \xi) + T_b(f, c, \eta), \quad (9.23)$$

where

$$C_0(t) = \frac{(t+d)(\gamma-2) - dA_0}{p-t} + \zeta(\gamma-2-A_0), \quad (9.24)$$

$$T_a = \left(1 + \frac{e}{a} \right)^a \prod_{i \in I} r_i \left(B_* - \sum_{i \in I} \frac{\xi_i}{r_i} \right) - \lambda_\xi S_\xi, \quad (9.25)$$

and T_b is the analogous column expression.

For a side of complement size a , group size $m = 6 - a$, support size k , and excess e , Lemma 9.3 gives

$$T_a \leq \left(1 + \frac{e}{a} \right)^a h^{k-1} \ell_0^{m-k} (B_* h - 1) - \lambda_\xi \left(\frac{1}{k} - \frac{1}{m} \right), \quad (9.26)$$

where h, ℓ_0 are given by (9.19) with $s = m - e$. The formula for $k = m$ omits the low level and entropy term. The column side is identical.

9.4 The exact four-variable certificate

For fixed $0 \leq t \leq 2E_0$, the enlarged constraints $-E_0 \leq e, f \leq E_0$ and $e + f = t$ are equivalent to $e \in [t - E_0, E_0]$ and $f = t - e$. Thus the following map covers that entire line segment

exactly; adjoining the boundary $t = 0$ only enlarges the actual feasible set, which has $t > 0$. Map this closed triangular domain to the unit square by

$$t = 2E_0x, \quad e = t - E_0 + z(2E_0 - t), \quad f = E_0 - z(2E_0 - t), \quad 0 \leq x, z \leq 1. \quad (9.27)$$

For every type (a, b) , every pair of support sizes

$$1 \leq k \leq 6 - a, \quad 1 \leq k' \leq 6 - b,$$

and the two spread parameters $u, v \in [0, 1]$, substitute (9.19), (9.27), and (9.21) into the right side of (9.23). Define

$$P_{a,b,k,k'}(x, z, u, v) = -(p - t)\mathcal{U}. \quad (9.28)$$

On the enlarged parameter cube,

$$p - t \geq p - \frac{11}{25} > 0, \quad A_0 > 0,$$

and every row and column denominator is at least $25/32$. Hence all rational expressions used to define $\mathcal{U}$ have positive denominators, and multiplication by $p - t$ reverses no inequality. Thus this is a rational polynomial. Lemma 8.8, implemented by exact tensor-product Bernstein subdivision, proves

$$P_{a,b,k,k'} > 0 \quad \text{on } [0, 1]^4 \quad (9.29)$$

for all 98 cases. The certificate summary is

(a, b)	polynomials	Bernstein nodes	maximum depth	
(1, 1)	25	165	6	(9.30)
(1, 2)	20	276	7	
(1, 3)	15	483	14	
(2, 2)	16	416	13	
(1, 4)	10	352	14	
(2, 3)	12	558	15	

Every terminal Bernstein coefficient is a strictly positive rational number. The smallest terminal coefficient for the tightest type $(2, 3)$ is

$$\frac{206812176444269171}{2388787200000000000000} > 0 \quad (\approx 8.657622 \times 10^{-5}). \quad (9.31)$$

The supplementary script constructs every polynomial from (9.28), performs exact power-to-Bernstein conversion, and uses exact de Casteljau subdivision at $1/2$.

Since $p - t > 0$, (9.29) gives $\mathcal{U} < 0$, contradicting

$$0 \leq F + \zeta G \leq \mathcal{U}.$$

Thus no two-sided critical Hall pair can occur.

9.5 One-sided critical Hall pairs

After transposition, assume $I = [6]$. As in Section 7, $b = 0$ would force zero Hall deficit, so $1 \leq b \leq 5$ and $|J| = p = 6 - b$. The complementary b columns have total mass $b + p\tau$; applying Lemma 9.1 to that positive excess gives $(p\tau)^2 \leq 15/319 < E_0^2$. Therefore

$$\sum_{j \in J} c_j = p(1 - \tau), \quad 0 < \tau < \frac{E_0}{p}. \quad (9.32)$$

The corresponding columns of B saturate $A/(1 - \tau)$. Put

$$m_i = \sum_{j \in J} b_{ij}, \quad \beta_i = \frac{m_i}{p}, \quad s = \sum_i \left(\beta_i - \frac{1}{6} \right)^2.$$

Lemma 7.3 gives

$$D := 1 - \sum_i \frac{\beta_i}{r_i} \geq \frac{(1 - \gamma)\tau}{1 - \tau} > 0. \quad (9.33)$$

Lemma 9.4 (Dimension-six row energy). *With $y_i = 1 - r_i^{-1}$, put*

$$C_6 = \frac{(1 - \gamma)(25/32)^2}{2}.$$

Then

$$\rho \geq C_6 \sum_i y_i^2, \quad C_6 > \frac{3}{10}. \quad (9.34)$$

In the present one-sided setting $D > 0$, so $y \neq 0$ and the comparison with $3/10$ is strict. Consequently,

$$\rho > \frac{3}{10} \frac{(1 - \gamma)^2 \tau^2}{(1 - \tau)^2 s}. \quad (9.35)$$

Proof. By Lemma 9.1, $r_i > 25/32$. The integral argument in Lemma 7.4 gives

$$r - 1 - \log r \geq \frac{(25/32)^2}{2} \left(1 - \frac{1}{r} \right)^2.$$

Since $1 - \rho \geq 1 - \gamma$,

$$\rho \geq (1 - \rho)(-\log(1 - \rho)) \geq \frac{(1 - \gamma)(25/32)^2}{2} \sum_i y_i^2.$$

The exact gap

$$C_6 - \frac{3}{10} = \frac{1547}{3317760} > 0 \quad (9.36)$$

proves the stated coefficient comparison. Since $D = \sum_i \beta_i y_i > 0$, y is nonzero. The harmonic-mean inequality gives $\sum_i y_i \leq 0$; Cauchy–Schwarz together with (9.33) then proves (9.35). In particular, $s > 0$. $\square$

Retain the exact complementary column-product loss. Two-group AM–GM gives

$$\sigma \geq 1 - Q_b(\tau), \quad Q_b(\tau) = \left(1 + \frac{p\tau}{b} \right)^b (1 - \tau)^p. \quad (9.37)$$

Because A is a boundary matrix, the dominated doubly stochastic matrix B has a zero. If $q^{(j)}$ are the p saturated columns, then convexity of squared distance gives

$$s \leq \frac{1}{p} \sum_{j \in J} s_6(q^{(j)}) \leq \max_{j \in J} s_6(q^{(j)}).$$

Apply Theorem 3.9 to the maximizing column and Corollary 3.6. Since $A \geq (1 - \tau)B$, we obtain

$$H \geq (1 - \tau)^6 \max\{\kappa, \gamma \exp(s/2)\}. \quad (9.38)$$

On the other hand,

$$H \leq \gamma - \rho - \sigma. \quad (9.39)$$

Set $s_0 = 2 \log(\kappa/\gamma)$.

If $s \leq s_0$, then $s_0 \leq 2(\kappa - \gamma)/\gamma$, and (9.35) gives

$$\rho > A \frac{\tau^2}{(1-\tau)^2}, \quad A = \frac{3}{10} \frac{(1-\gamma)^2 \gamma}{2(\kappa - \gamma)} = \frac{39750390625}{5253139008}. \quad (9.40)$$

It is therefore enough to prove

$$(1-\tau)^2 [1 - Q_b(\tau) + \kappa(1-\tau)^6 - \gamma] + A\tau^2 > 0. \quad (9.41)$$

For $b = 1, \dots, 5$, exact univariate Bernstein subdivision proves this on $0 \leq \tau \leq 11/(50p)$; the required depths are respectively 1, 2, 3, 4, 5.

If $s \geq s_0$, use $\exp(s/2) \geq 1 + s/2$ and AM–GM in (9.35). With $q = 1 - \tau$,

$$\rho + \gamma q^6 \exp(s/2) - \gamma > (1-\gamma)\tau q^2 \sqrt{\frac{3\gamma}{5}} + \gamma(q^6 - 1).$$

Here $3\gamma/5 = 1/108$, and

$$\frac{(1-\gamma)^2}{108} - \frac{9}{1024} = \frac{33853}{181398528} > 0. \quad (9.42)$$

Thus it remains to prove, for $\tau > 0$,

$$1 - Q_b(\tau) + \frac{3}{32}\tau(1-\tau)^2 + \gamma((1-\tau)^6 - 1) > 0. \quad (9.43)$$

The left side has a factor τ . For all five values of b , the quotient has globally positive Bernstein coefficients on $[0, 11/(50p)]$; the smallest coefficient is $1/864$.

Both entropy regimes contradict (9.38)–(9.39). Hence no one-sided critical pair exists.

Theorem 9.5 (Boundary exclusion in dimension six). *No global maximizer of Φ on K_6 has a zero entry.*

Proof. The case $\tau = 0$ was excluded after Lemma 9.1. If $\tau > 0$, a critical pair exists. The preceding subsections exclude both its two-sided and one-sided forms. Thus no boundary global maximizer exists. $\square$

10 Proof of the main theorem and corollaries

Proof of Theorem 2.1. Fix $6 \leq n \leq 15$.

- (1) Compactness of K_n and continuity of Φ give a global maximizer A .
- (2) Theorem 9.5 for $n = 6$, Theorem 8.10 for $n = 7$, Theorem 7.7 for $8 \leq n \leq 10$, and Theorem 6.4 for $11 \leq n \leq 15$ exclude every boundary maximizer. Thus A is positive.
- (3) The positive-maximizer theorem, Theorem 4.5, then gives $A = U_n$.
- (4) At U_n all row and column sums equal one and $\text{per}(U_n) = n!/n^n$, so

$$\max_{A \in K_n} \Phi(A) = \Phi(U_n) = 2 - \frac{n!}{n^n}.$$

Any matrix attaining this value is itself a global maximizer; repeating steps (2)–(3) proves that it is U_n . This establishes both the value and uniqueness. $\square$

Corollary 10.1 (No boundary maximizers). *For every $6 \leq n \leq 15$, every global maximizer of Φ lies in the relative interior of K_n .*

11 Discussion, scope, and verification status

The proof has several logically separate modules. Stable-polynomial capacity first yields the support-sensitive permanent inequality, the sharp rectangular-zero-block bound, and three entropy refinements. The exact complement entropy Ψ_m controls both a selected supported column and the weighted geometric mean of all complementary minors. These two forms combine into the core-marginal bound of Proposition 3.14.

The positive-maximizer theorem is independent of the boundary analysis. Pairwise averaging is flat at a positive global maximizer, while the Hessian at the uniform matrix is negative definite on the tangent hyperplane. Cyclic averaging therefore identifies the unique interior maximizer.

The common boundary framework consists of the joint deficit budget, relative-entropy subset estimates, Hall dilation, and support-sensitive permanent lower bounds. Scalar first- and second-order comparisons settle dimensions eight through fifteen and all ordinary dimension-seven cuts. The three dimension-seven single-bridge types require complementary-minor entropy, a core stationarity identity, and conditional product energy; the last separation is the exact two-variable alternative in Proposition 8.9.

In dimension six, marginal entropy must be retained even when the Hall core has size $p \geq 2$. The exact stationarity identity in Lemma 9.2 and the two-level reduction in Lemma 9.3 turn every two-sided configuration into one of 98 four-variable polynomial inequalities. The one-sided argument is sharper than its higher-dimensional counterpart because it keeps the full complementary column-product envelope $Q_b(\tau)$.

The computer-assisted component consists of finite exact rational certificates: scalar comparisons in all dimensions, the dimension-seven one- and two-variable Bernstein certificates, the 98 dimension-six four-variable certificates, and ten dimension-six univariate certificates. The analytic reductions, the polynomials being certified, the positivity criterion, and all subdivision statistics are stated in the text. No floating-point comparison, tolerance, random choice, or numerical optimizer is used.

The theorem established here is exactly the range $6 \leq n \leq 15$. We do not infer a complete classification of the remaining dimensions from the historical references: the cited $n = 4$ paper establishes full indecomposability of maximizers, while the $n = 16$ and $n \geq 17$ claims are recent preprints as of the version date. None of these external dimension claims is used in the proof above.

Meaning of “unconditional” and verification status. The proof does not assume any previously claimed dimension of Dittert’s conjecture, nor any other unproved conjecture. It does use standard results, among them compactness and continuity, weighted AM–GM, Cauchy–Schwarz, Bernoulli’s inequality, convexity, the harmonic-mean inequality, max-flow/min-cut, Hurwitz’s theorem, Gauss–Lucas, and finite-dimensional Taylor expansion. The finite comparisons are exact rational computations; the author checkers and independent reconstructions are included, and no success condition is implemented by a Python `assert`. A successful run therefore performs the same checks under ordinary and optimized Python execution. This is an exact computer-assisted proof in the usual mathematical sense, but it is not a Lean, Coq, Isabelle, or other proof-assistant-kernel formalization. Human refereeing of the analytic reductions remains essential.

Declarations

Code availability. All generators, exact checkers, independent audit implementations, tests, source hashes, and the machine-readable verification manifest are contained in the supplementary archive. The accepted version should additionally cite an immutable public repository release and archival DOI; those external identifiers cannot be assigned inside an anonymous review package.

Data availability. No external data set is used. Every finite datum required by the proof is generated from the displayed rational formulas and is included in the supplementary package.

Funding and conflicts of interest. No funding or conflict information is supplied in the anonymous version. The corresponding author must replace this sentence with the applicable journal declarations before acceptance.

Author contributions. The author(s) are responsible for the mathematical argument, the exact certificate software, and the final verification. A journal-specific CRediT statement should be inserted when the author list is disclosed.

Use of computational and language-assistance tools. Computer algebra and exact Python programs were used for formula expansion, certificate generation, checking, and manuscript preparation. Language and software-assistance tools were also used during revision. All mathematical claims, source transformations, and final outputs were reviewed and accepted by the author(s), who retain full responsibility for the work.

A Exact rational certificates

This appendix records representative exact comparisons used in the text. The supplementary programs verify every scalar margin and every Bernstein subdivision.

For dimension six,

$$\gamma_6 = \frac{5}{324}, \quad \kappa_6 = \frac{6144}{390625}, \quad \frac{15}{319} < \left(\frac{7}{32}\right)^2 < \left(\frac{11}{50}\right)^2.$$

The row-energy and large-entropy gaps are

$$\frac{(1 - \gamma_6)(25/32)^2}{2} - \frac{3}{10} = \frac{1547}{3317760}, \quad \frac{(1 - \gamma_6)^2}{108} - \frac{9}{1024} = \frac{33853}{181398528}.$$

The four-variable certificate is summarized in (9.30), and the smallest recorded terminal coefficient for its tightest type is (9.31). The one-sided certificates consist of five small-entropy polynomials and five large-entropy quotients; the latter have minimum Bernstein coefficient $1/864$.

For dimension seven, the localization and half-gap comparisons are

$$\eta_7^2 - \frac{\gamma_7}{14(1 - \gamma_7)} = \frac{23263}{1808073127}, \quad \frac{1}{2} - \frac{3}{7} - \eta_7 = \frac{33}{658}. \quad (\text{A.1})$$

The minimum ordinary two-sided margin is

$$\frac{217499955444827479495}{21713873952987016007066496},$$

and the minimum sign-mixed single-bridge margin is

$$\frac{165035926385}{3897251304526692}.$$

The one-sided energy, small-entropy, and large-entropy certificates are the positive quantities in (8.8), (8.10), and (8.13). The exact two-variable certificate is summarized in (8.38).

The remaining exact margins are displayed in equations (6.3), (6.13), (6.14), (7.5), (7.23), and (7.26).

For the localization constants in (5.9), one has

n	$\eta_n^2 - \frac{\gamma_n}{2n(1-\gamma_n)}$	$\frac{1}{2} - \frac{\lfloor (n-1)/2 \rfloor}{n} - \eta_n$
7	$\frac{23263}{1808073127}$	$\frac{33}{658}$
8	$\frac{4757}{836844800}$	$\frac{9}{80}$
9	$\frac{242489}{87087962025}$	$\frac{13}{270}$
10	$\frac{31109}{41313127850}$	$\frac{11}{115}$
11	$\frac{63007753811}{34945789841847500}$	$\frac{82}{1925}$
12	$\frac{11651173}{90828753405000}$	$\frac{319}{3900}$
13	$\frac{13072911571453}{366471344281458130000}$	$\frac{537}{14300}$
14	$\frac{71438752181}{2511343447066440000}$	$\frac{893}{12600}$

(A.2)

Every entry is positive.

The auxiliary comparison used for $11 \leq n \leq 14$ is

$$\frac{1}{1000} - \gamma_{11} = \frac{22308624601}{25937424601000} > 0. \quad (\text{A.3})$$

For $n = 8, 9, 10$, the inequalities $\gamma_n < 1/81$ follow from

n	$1/81 - \gamma_n$
8	$\frac{105557}{10616832}$
9	$\frac{54569}{4782969}$
10	$\frac{1516573}{126562500}$

(A.4)

For dimension 15, the six inequalities in (6.19)–(6.21) are equivalent to the positivity of the following integers:

$$\begin{aligned}
15^{15} - 300000 \cdot 15! &= 45\,591\,579\,980\,859\,375, \\
29863 \cdot 15^{15} - 10^{10} 15! &= 81\,568\,443\,603\,515\,625, \\
10^{10} 13! 13^{13} - 29937 \cdot 14^{26} &= 290\,584\,966\,207\,055\,614\,607\,247\,802\,368, \\
3 \cdot 14^{26} - 10^6 13! 13^{13} &= 3\,939\,835\,293\,455\,120\,947\,239\,518\,208, \\
10^8 13! 12^{24} - 301 \cdot 13^{37} &= 18\,198\,910\,159\,813\,162\,803\,331\,251\,792\,701\,065\,161\,767, \\
4 \cdot 13^{37} - 10^6 13! 12^{24} &= 162\,574\,843\,672\,041\,436\,633\,515\,011\,954\,046\,933\,353\,332.
\end{aligned} \quad (\text{A.5})$$

B Exact-arithmetic verification

The supplementary package contains two author verifiers, three independent reconstruction checkers, exact unit tests, and one orchestration script. The principal executable files are listed below.

File	Purpose
<code>ditttert_n6_certificates.py</code>	dimension-six generator and exact certificates;
<code>ditttert_n7_n15_certificates.py</code>	standard-library certificates for dimensions seven through fifteen;
<code>independent_n6_fast_audit.py</code>	independent reconstruction of all 98 four-variable certificates;
<code>independent_n6_univariate_audit.py</code>	independent standard-library reconstruction of the ten one-variable certificates;
<code>independent_n7_certificate_audit.py</code>	independent symbolic and Bernstein reconstruction of the three single-bridge trees;
<code>run_all_certificates.py</code>	environment-aware orchestration, hashing, manifest generation, and regression checks.

Run

```
python3 run_all_certificates.py.
```

A successful full run ends with

```
All Ditttert certificates for dimensions 6 through 15 passed.
```

It also writes the machine-readable files

```
verification_manifest.json and verification_output.txt.
```

Success is printed only after every subprocess exits normally and all expected statistics have been checked.

The dimension-seven-through-fifteen verifier requires Python 3.11 or newer and only the standard library. It evaluates every localization, Hall-range, first-order, and second-order margin as a `Fraction`; constructs the univariate and bivariate dimension-seven polynomials; performs exact power-to-Bernstein conversion and exact dyadic de Casteljau subdivision; and checks all reported tree statistics literally.

The dimension-six verifier was tested with Python 3.11–3.13; the archived reference environment uses Python 3.13, SymPy 1.14, and NumPy 2.3. SymPy is used only to expand the explicit rational polynomials, and NumPy only stores object arrays. Every generated coefficient is converted to `fractions.Fraction` before Bernstein conversion, subdivision, or comparison. The program constructs all 98 polynomials in (9.28), checks the node and depth counts in (9.30), and verifies the ten one-sided polynomials. The accompanying `requirements.txt` records the tested dependencies.

Neither verifier contains a floating-point comparison, tolerance, random choice, numerical optimization routine, or validation implemented by a Python `assert`. Ordinary and optimized Python execution therefore perform the same mathematical checks. The Bernstein implication is elementary: on a box, tensor-product Bernstein basis functions are nonnegative and sum to one, so a polynomial lies between the smallest and largest of its Bernstein coefficients.

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Textual object	Author-code entry point	Enumeration	Count
Dimension-six two-sided alternative, (9.28)	<code>two_sided_family,</code> <code>verify_two_sided</code>	six Hall types and all row/column support pairs	98
Dimension-six one-sided bounds	<code>verify_one_sided</code>	$b = 1, \dots, 5$, two entropy regimes	10
Dimension-seven single-bridge alternative, Proposition 8.9	<code>single_bridge_polynomials,</code> <code>certify_single_bridge</code>	$(a, b) = (1, 5), (2, 4), (3, 3)$	3 trees
Dimensions 8–10 second-order margins	<code>verify_n8_n10</code>	all admissible two-sided Hall types	$12 + 16 + 20$
Dimensions 11–14 first-order margins	<code>verify_n11_n14</code>	all listed Hall types, with one $n = 11$ refinement	$35 + 42 + 49 + 56$
Dimension 15 thin–thick bounds	<code>verify_n15</code>	two rational margins and six cleared-denominator tests	8

Table 2: Map from the analytic reductions to the exact verification code.

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